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Development of a Large Language Model-based Feature Extraction Pipeline to Leverage Clinical Notes for Interpretable Sepsis Prediction

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Summary

Sepsis remains one of the leading causes of mortality in intensive care units worldwide. Predictive models that can identify sepsis at an early stage therefore hold the potential to save lives, reduce the length of hospital stays, and improve patient outcomes. Traditionally, such models have relied mainly on structured electronic health record (EHR) data, such as laboratory values, vital signs, or demographic information. However, an often overlooked yet rich source of information is contained in unstructured clinical notes. These notes provide nuanced descriptions of symptoms, treatments, and clinical reasoning, but their integration into predictive models poses methodological challenges. Effective integration of clinical text into predictive models requires approaches that strike a careful balance between expressive power and interpretability. Methods must be sufficiently flexible to capture the heterogeneity and contextual richness of clinical documentation, while remaining transparent and robust enough to support clinical trust and downstream decision-making.

This thesis investigates methods that balance expressive power and interpretability when extracting features from free-text clinical documentation to improve sepsis prediction. To this end, three approaches were designed and compared. The first (M1) is a rule-based system that applies carefully crafted regular expressions and biomedical entity linking to detect mentions of sepsis-related symptoms and physiological parameters. The second (M2) employs a large language model (LLM) to classify symptoms probabilistically in a zero-shot setting, thereby leveraging their ability to interpret narrative context and implicit cues. The third (M3) is a hybrid approach that constrains LLMs with symbolic rules and prompts them to extract explicit numeric values, thus combining contextual understanding with structured precision. The comparative evaluation of these approaches highlights distinct strengths and weaknesses. The M1 method offered full transparency and low false positive rates, but it failed to recognize lexical variability or contextual descriptions. The M2 method provided high recall and excelled at interpreting implicit information, yet it often overgeneralized and produced false positives in ambiguous cases. The M3 method demonstrated superior precision by minimizing hallucinations, but its strict filtering reduces sensitivity and may lead to false negatives.

These methods were then applied to the MIMIC-III database and were integrated with structured variables and used as input to a Bidirectional Long Short-Term Memory (BiLSTM) model for sepsis prediction across several time horizons (0, 6, and 12 hours before onset). The benefit of clinical notes for sepsis prediction depends on extraction strategy and prediction horizon. At onset, text provides only marginal gains, with the hybrid approach (M3) performing best. At earlier horizons, LLM-based extraction (M2) substantially outperforms other methods by capturing higher-level clinical context, while rule-based extraction (M1) offers no benefit. Interpretability analyses reveal a trade-off between predictive performance and traceability, underscoring the need for horizon-aware text extraction strategies as interpretability is essential for clinical AI adoption, where decisions must be transparent, verifiable, and clinically aligned.

Abbreviations and Glossary

Artificial ventilation	Mechanical ventilation or respiratory support (SOFA-score)
ROC-AUC	Receiver Operating Characteristic - Area Under the Curve; a model performance metric
AUPRC	Area Under the Precision–Recall Curve
BiLSTM	Bidirectional Long Short-Term Memory network
Bilirubin	Indicator of liver dysfunction (SOFA-score)
Creatinine	Indicator of renal dysfunction (SOFA-score)
Dobutamine	Anti-hypotensive (SOFA-score)
Dopamine	Anti-hypotensive (SOFA-score)
EHR	Electronic Health Record
Epinephrine	Anti-hypotensive (SOFA-score)
Fever	Elevated body temperature (SIRS-criteria)
GCS	Glasgow Coma Scale, neurological assessment tool (SOFA-score)
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
Hypotension	Abnormally low blood pressure (SIRS-criteria)
Hypothermia	Abnormally low body temperature (SIRS-criteria)
ICU	Intensive Care Unit
Leukocytosis	Elevated white blood cell count (SIRS-criteria)
Leukopenia	Low white blood cell count (SIRS-criteria)
LLM	Large Language Model
MAP	Mean Arterial Pressure; blood pressure measurement (SOFA-score)
MIMIC-III	Medical Information Mart for Intensive Care III dataset
Norepinephrine	Anti-hypotensive (SOFA-score)
Oliguria	Urine output typically less than 500 mL per 24 hours (SOFA-score)

PaO_2/FiO_2	Oxygenation ratio (SOFA-score)
Platelets	Indicator of coagulation dysfunction (SOFA-score)
qSOFA	Quick Sequential Organ Failure Assessment
RAG	Retrieval-Augmented Generation; a LLM method with external retrieval
RegEx	Regular Expression, rule-based pattern matching method
Sepsis-3	Current international consensus definition of sepsis (2016)
SHAP	SHapley Additive exPlanations; a method for model interpretability
SIRS	Systemic Inflammatory Response Syndrome; early sepsis criteria
SOFA	Sequential Organ Failure Assessment score; assessment score for sepsis
Systolic Blood Pressure	Blood pressure measurement (qSOFA-score)
Tachycardia	Elevated heart rate (SIRS-criteria)
Tachypnea	Elevated respiratory rate (SIRS-criteria and qSOFA-score)
UMLS	Unified Medical Language System; ontology for medical concepts

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1. Introduction

1.1. Motivation

Sepsis is a severe and potentially fatal condition caused by the body's dysregulated response to infection [1]. It can lead to tissue damage, organ failure, and death if not recognized and treated promptly [1]. According to global health statistics, sepsis is responsible for a significant proportion of in-hospital mortality, particularly in Intensive Care Units (ICUs) [2]. The window of opportunity for effective treatment is often narrow, and delays in diagnosis can have fatal consequences [1]. Consequently, predictive models that can detect sepsis in its early stages have the potential to save lives, reduce the length of hospital stays, and improve the overall quality of care [3].

While many machine learning models have been developed to predict sepsis using structured data from Electronic Health Records (EHRs) such as lab results, vitals, and demographic information, these approaches often overlook an important source of information: clinical notes [3]. It is estimated that approximately 80% of all EHR data consists of this unstructured data [4] and contain contextual details on history, symptoms, treatments, and clinical reasoning not captured in structured fields [5]. With the rise of Natural Language Processing (NLP), crucial information embedded in clinical notes can now be incorporated into sepsis prediction models through text embeddings, substantially enhancing performance [4], [6], [7]. Yet evidence shows that while combining structured and unstructured data improves accuracy, most models remain opaque, with little interpretability or external validation [8].

Meanwhile, interpretability is essential in clinical AI. Black-box models, no matter how accurate, may be unsuitable for deployment in real-world medical settings if clinicians cannot understand, verify, or trust their outputs [9]. Models that fail to offer transparent reasoning are less likely to be adopted, even if they offer performance improvements [10]. Thus, interpretable models are not only a technical goal but a practical requirement for clinical integration.

At the same time, research landscape is shifting. Large-scale, de-identified datasets such as MIMIC-III enable reproducible sepsis studies [11], while Large Language Models (LLMs) are transforming text processing [12]. Unlike earlier approaches, modern LLMs can extract structured information directly from free-text with minimal manual annotation or task-specific finetuning, even reaching performance close to expert human annotators [13]. They recognize complex patterns in clinical language and, when guided by prompts, can adapt to diverse linguistic expressions without extensive preprocessing [12]. When used thoughtfully, LLMs can bridge the gap between free-text clinical documentation and structured predictive modeling. This

convergence of data availability, model capability, and clinical demand makes this an opportune moment to explore new ways of enhancing sepsis prediction through interpretable text-based methods.

1.2. Problem Statement

Despite growing interest in predictive sepsis modeling with unstructured clinical notes, current approaches face two central challenges: transparency and robustness.

First, many models rely on embeddings from pretrained language models such as ClinicalBERT or BioBERT [7], [14]. While these embeddings improve predictive performance, they operate on dense, opaque text representations that obscure the clinical factors driving a prediction. As a result, even accurate models provide little insight into why a prediction was made, limiting their clinical utility [9].

Second, clinical notes themselves are highly variable in vocabulary and formatting. A single lab test such as *Creatinine* may appear as `Creatinine`, `Kre`, `K`, `Cr (mg/dL)`, or `Kreatinine: 1.4`. Such heterogeneity makes simple string matching unreliable, particularly when context or units are essential for interpretation. Addressing this variability requires more robust extraction methods.

This thesis argues that a more interpretable approach to clinical text modeling is both feasible and necessary. It addresses a key limitation in the current landscape of clinical AI: the absence of transparent, text-aware prediction pipelines that can reliably support real-world clinical decision-making. To explore the trade-off between interpretability and contextual understanding, this work introduces two novel LLM-based extraction methods.

The first method employs Retrieval-Augmented Generation (RAG) to identify relevant clinical notes and subsequently queries the LLM to estimate the probability that a note contains evidence of a specific symptom. The second method adopts a more constrained strategy by pre-selecting text segments using predefined symptom- or variable-specific strings, after which an LLM extracts the corresponding measurement values from the surrounding context. This design enables the recovery of clinically meaningful features even when values are expressed in heterogeneous or non-canonical forms, while preserving interpretability through rule-based anchoring.

Both approaches are evaluated against a purely rule-based baseline to quantify the benefits of incorporating contextual information into interpretable clinical text extraction.

1.3. Objective

The overall objective of this thesis is to develop an interpretable framework for enhancing sepsis prediction by integrating structured EHR data (MIMIC-III) with information extracted from unstructured clinical notes. The focus is on identifying explicit, clinically relevant parameters within free-text documentation and incorpo-

rating them into predictive models in a transparent manner. This work does not aim to produce a fully optimized or clinically deployable sepsis prediction system; instead, it evaluates the added value of text-derived features within a basic predictive setup.

The research questions guiding this work are as follows:

1. How accurately can regular expressions, LLMs, and hybrid models extract sepsis-related clinical features from unstructured clinical notes? (Section 5)

2. How do features extracted by rule-based, LLM, and hybrid methods differentially affect the performance and interpretability of structured sepsis prediction models? (Section 7)

1.4. Contribution

This thesis contributes to the field of interpretable clinical machine learning through the following innovations:

- Three pipelines to extract features based on a rule-based, an LLM-based, and a novel hybrid approach that combines regular expressions with LLM prompting to identify interpretable clinical parameters from unstructured notes.

- An integrative modeling approach that merges structured EHR data with explicitly extracted text features to enhance sepsis prediction while preserving transparency as well as an empirical evaluation using the MIMIC-III dataset to assess the effectiveness and interpretability of the proposed approach compared to baseline methods.

- Provision of reproducible code and workflows to ensure transparency and facilitate future research.

All code, preprocessing scripts, and experiment configurations used in this thesis are accessible and described in Appendix C, supporting reproducibility and future research.

1.5. Thesis Outline

This thesis is structured into eight chapters, each building toward the development of an interpretable sepsis prediction pipeline using LLMs and hybrid feature extraction strategies.

The opening chapters establish the conceptual and contextual foundations of the study. The current chapter (Chapter 1) introduces the motivation behind the study, outlines the central research questions, and presents the objectives and contributions of this work in the context of sepsis prediction using clinical notes. Chapter 2 provides a comprehensive review of relevant literature, including traditional rule-based methods and LLMs for clinical information extraction, the challenges of sepsis

diagnosis, predictive modeling approaches, and the role of explainable AI in clinical contexts. It concludes by identifying current gaps in the literature that this thesis will address. Chapter 3 describes the MIMIC-III dataset used in the study, detailing its composition, preprocessing requirements, and limitations in terms of generalizability, data quality, and temporal relevance.

The first part of the research focuses on feature extraction from clinical notes. Chapter 4 outlines the three methods developed to extract clinical features from unstructured notes. First, a rule-based approach using regular expressions and entity linking (M1). Second, a LLM-based probabilistic classification using zero-shot prompts and retrieval-augmented generation (M2). Lastly, a hybrid method that guides LLMs to extract structured numeric values based on regex and UMLS matches (M3). Chapter 5 evaluates the performance of the three feature extraction methods across metrics such as coverage, granularity, and interpretability. Strengths, limitations, and trade-offs are discussed for each method, along with proxy metrics and expert validation.

The second part of the research focuses on integrating the extracted features into an explainable sepsis prediction model. Chapter 6 describes the construction of a BiLSTM-based time-series prediction model that integrates structured EHR data with features extracted from clinical text. It details the preprocessing pipeline, model architecture, and the SHAP-based interpretability approach. Chapter 7 assesses the predictive performance of the BiLSTM model using no notes, M1, M2, and M3 across three prediction horizons (0, 6, and 12 hours). This chapter further presents SHAP-based feature attribution analyses to examine model interpretability, highlighting the clinical relevance of the most influential features and the trade-offs between predictive performance and transparency across the different extraction approaches.

Finally, Chapter 8 summarizes the key contributions of the thesis, answers the research questions, reflects on limitations, and outlines directions for future work, such as improving temporal reasoning and extending methods to real-time clinical settings.

2. Related Work

This chapter reviews prior work on leveraging unstructured clinical text for sepsis prediction. Section 2.1 introduces Electronic Health Records (EHRs) and their inconsistencies between structured and unstructured formats. Section 2.2 discusses rule-based methods for extracting information from unstructured formats, whereas Section 2.3 discusses Large Language Models (LLMs). Section 2.4 covers approaches for adapting LLMs to clinical tasks. Section 2.5 discusses the diagnostic challenges of sepsis and its evolving definitions. Section 2.6 surveys predictive modeling strategies. Section 2.7 reviews Explainable AI (XAI) methods that enhance transparency for clinicians. Finally, Section 2.8 outlines key limitations in current research, particularly the trade-off between performance and interpretability.

2.1. Electronic Health Records

Medical Data Each hospital stay generates a vast amount of information, ranging from administrative details to vital sign recordings such as blood pressure, temperature, and heart rate. Historically, this information was recorded manually, resulting in inefficiencies, fragmented records, and limited reusability for downstream analysis. The emergence of EHRs has transformed healthcare data management by enabling the systematic digital capture of patient information across clinical settings [15]. EHR data encompasses a wide range of patient information, including identifiers, prescribed medications, prior diagnoses, and vital signs [16].

While initially designed for clinical documentation and administrative tasks, EHR have become valuable for medical research as they can provide medical data for tasks such as disease prediction [3], [5]. Nevertheless, some challenges persist when using this data for medical research. Privacy regulations such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union impose strict restrictions on access to EHR data [17], [18]. To protect patient confidentiality, EHR datasets must be de-identified before use. As the risk of re-identification persists, especially when EHR data are combined with external data sources [19], access to publicly available EHR datasets often requires users to undergo certification [20].

Furthermore, data quality remains a key issue in both structured and unstructured formats. Inconsistent data entry practices, missing values, duplicate records, and heterogeneity in documentation across institutions all degrade the reliability of downstream analysis [5]. One survey found that 18% of healthcare records were duplicates, pointing to widespread issues in data integration and redundancy [21].

Structured vs. Unstructured Data EHR data is typically divided into two main categories: structured and unstructured data [5]. Structured data includes predefined, machine-readable elements such as lab values, medication codes, timestamps, and diagnostic labels [16]. These data types are easier to extract and analyze using traditional statistical or machine learning approaches. In contrast, unstructured data comprises free-text clinical narratives such as nursing and physician notes, discharge summaries, procedures, imaging, and pathology notes [5]. While rich in clinical detail, these documents are harder to process due to variability in writing style, medical abbreviations, and contextual nuances [5].

In order to address this variability, standardized medical vocabularies are introduced which provide a shared, structured language that reduces documentation variability and enables accurate interpretation of EHR data [22]. Several controlled vocabularies form the backbone of modern EHR systems, such as RxNorm for medications, LOINC for laboratory tests, and SNOMED CT for clinical concepts [23]–[25]. Despite these standards, many institutions continue to rely on heterogeneous or proprietary coding systems, creating semantic inconsistencies that hinder data interoperability [19]. The Unified Medical Language System (UMLS) addresses this by integrating multiple vocabularies into a single standardized ontology [26]. However, UMLS's complexity and inconsistent granularity can make implementation difficult [27]. To further promote standardization, the OMOP Common Data Model (CDM) was introduced, organizing over 10 million clinical concepts to support large-scale, multi-site analysis [28]. While these efforts have significantly improved the consistency of structured data, a large portion of clinically relevant information remains in unstructured form, which falls outside the scope of these controlled vocabularies and models.

Limitations of EHR In summary, while the adoption of EHRs has transformed healthcare by enabling the systematic digital capture of patient data, their use in medical research presents several challenges. Issues related to data quality, heterogeneity, and privacy constraints complicate the extraction of reliable insights, particularly from unstructured clinical text. Although standardization efforts have improved the usability of structured data, a substantial portion of clinically relevant information remains embedded in free-text documentation. Developing robust and interpretable methods to leverage this unstructured data therefore represents a key opportunity for improving data-driven clinical decision support, including early sepsis prediction.

2.2. Rule-Based Approaches

Rule-Based Information Extraction Rule-based approaches are Natural Language Processing (NLP) techniques that use pre-defined patterns and rules to extract certain concepts from the text [29]. Prior to the widespread adoption of machine learning models, rule-based systems were preferred for their interpretability and effective use of domain expertise [30].

Rule-based information extraction typically follows two strategies: pattern-based extraction, commonly implemented using regular expressions, and terminology-based mapping, which leverages clinical ontologies such as the Unified Medical Language System (UMLS) to identify and normalize medical concepts [31], [32]. Hybrid pipelines that combine these strategies can improve coverage and robustness, particularly in the presence of informal or heterogeneous clinical language [33]. Many implementations rely on established NLP frameworks to convert unstructured clinical narratives into structured representations [30], [34].

Many rule-based clinical NLP pipelines rely on established biomedical text processing frameworks that support named entity recognition and ontology-based concept linking, often to resources such as UMLS [35]. While these frameworks may internally employ neural models for entity recognition, they are commonly integrated into rule-based systems due to their standardized outputs and support for terminology mapping. However, approaches that depend on predefined vocabularies and standardized resources often lack flexibility, limiting their ability to adapt to institution-specific documentation styles or emerging terminology [34].

In contrast, pattern-based extraction using regular expressions can be tailored to the idiosyncrasies of a specific dataset or clinical setting. Such methods are particularly effective for identifying recurring and lexically predictable patterns (e.g., "BP: 120/80", "breathing fast") and are highly interpretable [30]. Nevertheless, they require substantial manual effort to develop and maintain and typically generalize poorly across institutions and datasets [34].

Limitations of Rule-Based Approaches Rule-based methods offer precision and transparency, making them attractive in clinical settings where expert oversight and traceability are essential [9]. However, their reliance on explicitly defined lexical patterns and predefined concepts fundamentally limits their flexibility and scalability. Small variations in wording, documentation style, or clinical phrasing can prevent successful extraction, and adapting rule sets to new institutions or datasets requires substantial manual effort [30], [34]. More importantly, rule-based systems struggle to capture subtle, implicit, or context-dependent clinical signals that are not expressed in standardized or predictable forms. In the context of sepsis, early warning signs such as vague symptom descriptions, evolving changes in mental status, or expressions of clinician concern are often documented indirectly in free-text notes, rendering them difficult to detect using rigid rule-based logic [36], [37]. As a result, while rule-based approaches remain effective for narrow and well-defined extraction tasks, their performance degrades in large-scale or heterogeneous clinical datasets.

Methodologically, rule-based and LLM-based extraction differ not only in expressive capacity but also in their failure modes. Rule-based systems tend to fail silently by omitting relevant information when expected patterns are absent, whereas LLM-based approaches may generate plausible but unsupported inferences [30], [38]. Accordingly, rule-based pipelines require manual rule engineering, whereas LLM-based approaches focus on prompt design and output validation [32], [34], [39].

2.3. Large Language Models

Representation Learning with Transformers Language models introduce a fundamentally different approach to processing clinical text by learning dense, contextual representations that preserve both semantic meaning and sequential structure [40], [41]. These document-level embeddings enable downstream models to exploit unstructured clinical notes more effectively, improving predictive performance in tasks such as early sepsis detection [41]. Prior to transformer-based models, approaches such as topic modeling with Latent Dirichlet Allocation (LDA) and vector-based techniques like TF–IDF were commonly used, but these methods produce sparse or shallow representations that fail to capture long-range dependencies and contextual nuance [4], [42], [43].

Introduced by Vaswani et al. (2017), the Transformer marked a paradigm shift in NLP by modeling long-range dependencies with high efficiency [44]. The original encoder–decoder design generates contextual embeddings that form the basis for both understanding and generation of text. Encoder-only adaptations produce bidirectional contextual embeddings for tasks like classification, retrieval, and similarity search [7], [42], [45]. To adapt such models to the clinical domain, variants such as ClinicalBERT were pretrained on large-scale clinical note corpora, enabling them to capture EHR-specific language and abbreviations [46].

LLMs for Clinical Text Built on decoder-only or encoder–decoder Transformers, modern LLMs extend NLP capabilities by synthesizing structured outputs from raw clinical text. This makes them powerful tools for annotation, where they efficiently extract relevant details and can approach the accuracy of human experts [13]. In sepsis prediction, this ability has been leveraged to summarize notes into structured forms that, once embedded with bidirectional contextual embeddings, enhance predictive performance [7]. Compared to traditional rule-based systems, LLMs better accommodate linguistic variability and contextual subtleties, reducing manual preprocessing while improving generalizability.

Opacity and Verifiability of LLM Outputs LLMs are fundamentally opaque, as their internal representations and decision-making processes are not directly interpretable [47]. This opacity poses a challenge for clinical decision support, where traceability and justification of model outputs are essential for clinician trust and accountability [9]. A direct consequence of this limited transparency is the difficulty in verifying whether generated outputs are grounded in clinical evidence or reflect unsupported inferences, including hallucinated content [38]. As a result, understanding and constraining how LLMs transform clinical text into actionable information remains a central challenge for their safe use in healthcare.

2.4. Adapting LLMs for Clinical Tasks

Challenges in Clinical Deployment Although LLMs demonstrate strong general-purpose language understanding capabilities [12], their deployment in clinical settings presents distinct challenges. Medical text is highly specialized, context-dependent, and often incomplete, requiring precise and conservative interpretation [48]. In this setting, hallucinated or overconfident outputs can directly compromise patient safety by influencing incorrect diagnoses or treatments [38]. Consequently, LLMs must be carefully adapted to specific clinical tasks through constraint mechanisms, domain-specific prompting, or structured input representations to ensure safe and reliable behavior.

Retrieval-Augmented Generation One strategy to address hallucinations in LLMs is RAG, which integrates a retrieval component with a generative model [49]. The retriever identifies relevant passages from external sources, such as medical literature, which are then provided as context for the generator to produce responses. This architecture has shown promise in clinical applications, where grounding responses in evidence is critical [12]. By incorporating structured or semi-structured knowledge during inference, RAG helps mitigate hallucinations and improve relevance and reliability in clinical outputs.

Prompting and Knowledge-Guided Adaptation Another strategy to reduce hallucinations is zero- and few-shot learning. With zero-shot learning LLMs rely solely on carefully designed instructions to generate task-specific outputs as well-crafted prompts can constrain the model and reduce unsupported generations [39], [50]. Few-shot learning builds on this by including example-based prompts that further guide the model toward accurate reasoning and improved outputs, even with limited task-specific data. Both approaches support rapid adaptation to clinical information extraction tasks, but their effectiveness is limited by the complexity of medical language, including abbreviations, polysemy, and inconsistent formatting [32]. The last strategy to improve accuracy and reduce hallucination is to incorporate biomedical knowledge during inference. Rule-based systems like UMLS MetaMap [31] and lexical normalization tools can be used to map clinical text to structured concepts. Bhattarai et al. showed that enriching LLM prompts with UMLS-derived terms improved clinical entity extraction accuracy and outperforming RAG-style pipelines in certain tasks [32]. This integration enables models to align free-text language with established ontologies and reduces ambiguity.

Limitations of Adapted LLMs Despite advances in adaptation strategies such as retrieval-augmented generation, prompt engineering, and the incorporation of external medical knowledge, LLMs remain imperfect tools for clinical deployment. In particular, LLMs are still susceptible to hallucinations when clinical documentation is sparse, ambiguous, or incomplete [48]. This limitation is especially critical in early

sepsis prediction, where relevant signals may be subtle, temporally dispersed, or only implicitly documented [1]. Even when grounded with retrieved context or structured knowledge, LLM outputs can reflect overconfident or unsupported inferences, posing risks for clinical decision support [38]. These challenges motivate the use of LLMs in constrained roles, such as structured feature extraction, rather than as end-to-end predictive or decision-making systems.

2.5. Challenges in Understanding Sepsis

Clinical Significance and Global Burden Sepsis is defined as "a life-threatening organ dysfunction due to a dysregulated host response to infection" [1]. It remains a critical global health concern, with an estimated 49 million cases and 11 million deaths reported in 2017, accounting for nearly 20% of all global deaths [2]. Early detection and treatment are essential, given the rapid progression and high mortality associated with the condition.

Diagnostic Criteria and Clinical Ambiguity Definitions of sepsis have shifted as understanding of its pathophysiology has advanced. Terms such as "sepsis syndrome" are now outdated, and distinctions like "sepsis" versus "severe sepsis" were often inconsistently applied [1]. Variation in ICD coding strategies has further complicated reliable case identification, hindering both clinical reporting and research comparability. Historically, diagnosis was guided by Systemic Inflammatory Response Syndrome (SIRS) criteria (Table A.2 on page ii) [51], which relied on abnormalities in temperature, heart rate, or WBC count. However, these markers lacked specificity, as they are common among hospitalized patients without infection [1].

To address these shortcomings, the 2016 Sepsis-3 guidelines introduced the most recent SOFA score, which evaluates dysfunction across six organ systems: respiratory, cardiovascular, coagulation, liver, CNS, and renal (Table A.1 on page i) [52]. A change of two or more points from baseline in a patient with suspected infection is considered diagnostic for sepsis [1]. When full SOFA parameters are unavailable, the simplified qSOFA tool (Table A.3 on page ii) can be applied at the bedside and is based on altered mentation (Glasgow Coma Scale < 15), systolic blood pressure ($\leq$ 100 mmHg), and respiratory rate ($\geq$ 22/min) [52]. Although convenient, qSOFA has limited sensitivity and may fail to capture many early sepsis cases, underscoring the need for additional diagnostic support [36].

Time Sensitivity and Diagnostic Barriers Despite improved definitions, early recognition of sepsis remains difficult due to its non-specific and variable presentation [1], [36]. Symptoms often overlap with other conditions, and reliance on scoring tools can delay diagnosis, especially in fast-paced or resource-constrained environments. Each hour of delay in antibiotic administration significantly increases mortality risk [53]. Even among survivors, long-term consequences such as cognitive decline and reduced quality of life underscore the urgency of timely intervention [54].

2.6. Predictive Modeling Approaches for Sepsis

Structured Data and Machine Learning Models Many predictive models for sepsis rely on structured EHR data such as vital signs and laboratory results. Early work by Futoma et al. [55] demonstrated that recurrent neural networks applied to multivariate physiological time series can anticipate sepsis onset hours before clinical recognition, establishing deep temporal models as a viable approach for early warning systems. Clinical scores like SOFA remain widely used to quantify organ dysfunction [4], but they offer only limited predictive value [4]. ML approaches have shown promise in anticipating sepsis onset hours before clinical recognition [6], [43], [56]–[58], potentially providing a critical window for intervention. Common architectures include recurrent networks such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU)[59], as well as convolutional and temporal convolutional networks (TCNs) [60]. A notable example is the MGP-TCN model, which combines a Gaussian process to handle irregular sampling with a TCN for sequence modeling, enabling accurate end-to-end prediction of sepsis risk [61]. Despite strong performance, models relying solely on structured variables fail to capture narrative clinical context documented in free-text notes, motivating the integration of unstructured data.

Unstructured Data and Natural Language Processing Alongside structured signals, unstructured clinical notes capture rich contextual information that may aid early recognition. Initial approaches applied traditional NLP methods such as TF–IDF, bag-of-words, and Latent Dirichlet Allocation (LDA) [4], [43], [56], [58], [62], [63]. For example, Goh et al. used LDA to cluster topics in notes, incorporating patient symptoms and social history to improve prediction [4].

While informative, such models reduce language to shallow statistical patterns and fail to capture nuanced clinical dependencies. Although feature weights or topic distributions are inspectable, predictions cannot be traced back to specific text segments, limiting auditability and verification of model behavior [9].

Transformer Models and Large Language Models Recent advances in transformer architectures have shifted focus toward LLMs, which offer deeper contextual understanding. Li et al. combined GPT-3.5-Turbo for summarization with Clinical-BERT embeddings for sepsis prediction, outperforming traditional NLP methods [7]. However, embedding-based approaches tend to capture information at the global document level, missing localized clinical cues, and remain limited by domain shifts in documentation and interpretability challenges [37].

To address these limitations, Shashikumar et al. proposed an LLM-enhanced system built on top of a pre-existing model, COMPOSER [37]. When the risk estimate from structured data was outside of a certain range, the system invoked a retrieval-augmented generation (RAG) pipeline: relevant clinical notes were first retrieved and ranked by similarity to sepsis-related queries, after which the LLM extracted key textual features (e.g., altered mentation or a recent positive COVID

test) to refine the prediction. Deployed in real-time, this hybrid system achieved strong predictive accuracy and demonstrated clinical utility, and succesfully reduce the false alarm rate while maintaining similar sensitivity levels.

Limitations Despite substantial progress, existing approaches have not fully leveraged the representational power of LLMs while prioritizing interpretability in clinical prediction. While ML and LLM-based systems improve early detection, they often operate as opaque black boxes [9]. In high-stakes settings like sepsis, clinicians must understand the basis of predictions to trust and act on them [9].

2.7. Explainable Artificial Intelligence in Medical Context

The Need for Transparency in Healthcare AI Improving transparency is central to the safe use of AI in medicine. LLMs and other advanced systems lack clarity in how predictions are generated [40]. This "black-box" nature raises particular concerns in high-stakes clinical domains, where interpretability is a prerequisite for trust and accountability [9]. By clarifying how outputs are derived, interpretability techniques enable clinicians to validate predictions, identify errors, and incorporate model insights into their workflow, ensuring a safer and more ethical AI adoption [64]. Regulatory frameworks echo this demand. The European Union's AI Act, for example, mandates that AI systems, particularly in sensitive fields like healthcare, must be "human-centric and trustworthy" [65], [66]. This underscores the dual requirement that clinical AI be both accurate and interpretable.

Foundations of XAI To address these challenges, the field of XAI has emerged with the goal of making AI systems understandable [64]. Broadly, XAI methods fall into two categories: intrinsically interpretable models (e.g., decision trees, linear regression) and post-hoc explanation techniques, which provide interpretability for complex black-box models [67]. The latter are especially relevant in healthcare, where neural networks and ensemble methods are often used for predictive modeling. SHAP (SHapley Additive exPlanations) is widely adopted post-hoc methods. Based on cooperative game theory, SHAP attributes importance to input features, offering both local (individual prediction) and global (dataset-level) explanations [68]. This dual perspective makes it particularly useful for structured clinical data such as vital signs, laboratory results, and diagnostic codes. Clinically, SHAP helps stakeholders identify which features drive predictions, enabling real-time validation and improving trust in AI-supported decisions. Compared to other methods, SHAP provides greater consistency and stability across model runs, though it can be computationally intensive [64]. Despite this limitation, it has become a cornerstone of interpretable machine learning in medicine.

XAI in Sepsis Onset Prediction Explainable models have been increasingly applied to sepsis prediction tasks [67]. Ensemble algorithms such as Random Forest and XGBoost are commonly paired with SHAP to identify key predictors, consistently highlighting vital signs such as heart rate, respiratory rate, and body temperature which align with clinical expectations [67]. These explainable prediction models support early intervention, strengthen clinical confidence and remove bias [67].

Balancing Performance and Interpretability in Sepsis Prediction By improving transparency, XAI techniques such as SHAP aim to bridge the gap between algorithmic performance and clinical usability [64]. In the context of sepsis prediction, this is particularly important, as clinicians must act rapidly while also understanding and validating the basis of AI-generated alerts. Despite substantial advances, existing approaches continue to exhibit a fundamental trade-off between predictive performance and interpretability. Rule-based systems and clinically grounded scoring models offer transparency and traceability, but their limited expressiveness restricts their ability to capture complex and context-dependent signals present in unstructured clinical text. In contrast, machine learning and LLM–based approaches achieve improved predictive performance by leveraging high-dimensional representations, yet often operate as opaque black boxes whose outputs are difficult to verify or justify in clinical practice [9]. In high-stakes settings such as sepsis care, this lack of transparency poses a barrier to clinical trust, regulatory acceptance, and real-world adoption.

Recent work has demonstrated that LLMs can be guided to produce structured and clinically grounded outputs through prompt design and task-specific constraints. Frameworks such as CHiLL explicitly extract interpretable features from clinical text in an auditable manner [39]. However, this principle has not yet been fully realized in sepsis prediction. For example, Shashikumar et al. [37] demonstrate the prospective deployment of an LLM-enhanced sepsis prediction system, but report predictions primarily as a continuous risk score without exposing the underlying textual features contributing to model outputs. As a result, while improved performance is shown, the specific clinical concepts or narrative cues driving predictions remain opaque, limiting interpretability and the ability for clinicians to assess, validate, or contest individual alerts.

2.8. Limitations

Despite significant advances in sepsis prediction, existing approaches have not yet fully reconciled the need for strong predictive performance with the requirements of transparency, verifiability, and clinical trust. Across the literature, approaches to sepsis prediction using clinical text have been end-to-end black-box pipelines [3], [41], leaving limited exploration of hybrid designs that decouple feature extraction from prediction.

As a result, there remains a clear research gap in developing approaches that leverage the contextual strengths of LLMs while maintaining transparency, verifiabil-

ity, and clinical reliability. This gap is especially critical for early sepsis detection, where subtle and context-dependent signals must be extracted reliably from free-text clinical documentation. This thesis addresses this gap by investigating constrained, interpretable feature extraction from unstructured clinical notes and integrating these features with structured EHR data to support transparent and trustworthy early sepsis prediction.

3. MIMIC-III Dataset

Medical Information Mart for Intensive Care III (MIMIC III) is a publicly available, de-identified dataset comprising clinical data from over 40,000 Intensive Care Unit (ICU) patients admitted to Beth Israel Deaconess Medical Center between 2001 and 2012 [11], [69], [70]. Developed by the MIT Laboratory for Computational Physiology, it includes detailed information such as patient demographics, physiological measurements, medications, laboratory test results, procedures, and unstructured clinical notes.

The dataset is HIPAA-compliant and de-identified. Researchers gain access to the dataset after completing human-subjects training (e.g., CITI) and signing a data use agreement.

MIMIC-III has become a cornerstone for machine learning and clinical informatics studies. It has enabled research in diverse areas, including in-hospital mortality prediction, early detection of sepsis, and benchmarking of clinical decision support systems [7], [39], [71]. Consequently, methodologies developed using MIMIC-III can be directly compared against a wide array of established approaches.

3.1. Overview of MIMIC-III Data

The MIMIC-III database is organized into 26 relational tables containing both structured and unstructured information [70]. The data were originally collected during routine hospital care at the Beth Israel Deaconess Medical Center (BIDMC) and compiled from several institutional sources, including archives of ICU information systems, hospital electronic health records, and the Social Security Administration Death Master File for mortality follow-up.

A key aspect of MIMIC-III is that it integrates data from two different critical care information systems used at BIDMC over the data collection period: CareVue, covering ICU stays between 2001–2008, and Metavision, covering ICU stays between 2008–2012.

The data from both systems were merged when compatible. If not, separate tables were kept for differing structures (notably for fluid intake), distinguished by suffixes such as `INPUTEVENTS_CV` and `INPUTEVENTS_MV`. Data quality also differs between the systems, with MetaVision providing more detailed information on procedures and medication administration than CareVue [72].

Core identifiers Three key tables organize patient-level information in MIMIC-III. `PATIENTS` records demographic data such as age, sex, and date of death; `ADMISSIONS` captures hospital-level details including admission and discharge times, insurance

type, ethnicity, and primary diagnosis; and ICUSTAYS contains ICU-specific metadata such as unit type, admission and discharge times, and a unique identifier for each stay. Together, these tables link demographic, hospital, and ICU information, forming the foundation for cohort construction and temporal alignment of events across the database.

Structured clinical data Structured data span bedside observations (CHARTEVENTS), laboratory tests (LABEVENTS), microbiology results (MICROBIOLOGYEVENTS), medication and fluid records (INPUTEVENTS_CV/MV, OUTPUTEVENTS), and coding information (DIAGNOSES_ICD, PROCEDURES_ICD, PRESCRIPTIONS). Supporting tables (D_ITEMS, D_LABITEMS) map numeric identifiers to readable labels and units. Together, these tables form the structured backbone of the database, enabling detailed reconstruction of patient physiology, treatments, and diagnoses throughout the ICU stay.

Unstructured clinical notes The NOTEEVENTS table contains over two million free-text notes covering diverse aspects of ICU care. Major categories include *nursing notes* (bedside observations and shift reports), *physician notes* (daily assessments and treatment plans), and *discharge summaries* (retrospective overviews of diagnoses and outcomes). Additional documentation includes specialized reports such as *Radiology*, *ECG*, and *Echocardiography*, as well as notes from *Respiratory therapy*, *Nutrition*, *Pharmacy*, and *Social work*. Together, these texts provide complementary narrative context that extends beyond structured measurements.

Relevant tables For the purposes of this thesis, only a subset of the MIMIC-III tables was utilized. To ensure comparability of cohort extraction and sepsis label definition, all tables accessed in the study of Moor et al. [61] were also employed. These include PATIENTS, ADMISSIONS, and ICUSTAYS for demographic and cohort definitions; CHARTEVENTS, LABEVENTS, OUTPUTEVENTS, INPUTEVENTS_MV, and INPUTEVENTS_CV for structured clinical observations; PRESCRIPTIONS and MICROBIOLOGYEVENTS for medication orders and microbiological evidence of infection; and the dictionary tables D_ITEMS and D_LABITEMS for mapping item identifiers to standardized labels and units.

In addition, the NOTEEVENTS table was accessed, restricted to *Physician* and *Nursing Notes*.

3.2. Limitations of MIMIC-III

Despite its broad utility, MIMIC-III exhibits several important limitations related to population representation, data quality, and generalizability.

Population and Representation Bias First, MIMIC-III is a single-center dataset which limits institutional and geographic diversity, potentially biasing the dataset

toward specific demographic groups. For example, the dataset overrepresents older, white, and insured patients, underrepresenting racial minorities and those without regular access to tertiary care [70], [73]. These limitations in population representativeness have been shown to affect the fairness and generalizability of machine learning models trained on MIMIC-III data, particularly in terms of demographic subgroup performance and reproducibility across healthcare systems [73].

Dated Coding Systems Second, MIMIC-III spans the years 2001 to 2012, during which clinical guidelines and ICU practices have undergone substantial change. The reliance on outdated coding systems such as ICD-9 further hinders integration with modern datasets using ICD-10 or SNOMED CT [74]. As a result, predictive models trained on MIMIC-III may suffer from temporal drift and limited contemporary clinical relevance.

Data quality Third, variable completeness, irregular sampling (minute-level vitals vs. hourly labs), inconsistent coding, and unit mismatches complicate data integration. Timestamps such as `charttime` vs. `storetime` often diverge, challenging temporal analysis [70]. Clinical notes are also irregularly logged with fewer than 9% of ICU admissions include physician progress notes [75]. Additionally, as a single-center dataset, MIMIC-III reflects local clinical workflow and documentation practices, constraining external generalizability [76]

De-identification Constraints Fourth, while the dataset is de-identified to comply with HIPAA, key clinical information is altered or omitted in ways that may influence research. Dates are shifted with each patient receiving a unique offset placing all events into the years 2050 - 2200, preserving intervals but removing real calendar context and precluding seasonality or historical trend analyses [70]. Additionally, free-text notes are automatically scrubbed of all Personal Health Information such as names and dates using pattern-matching and dictionary methods, which may remove clinically relevant context [70]. Patients over age 89 have their ages set above 300 years, which prevents precise age-based cohort modeling in older populations [70].

Access and Reproducibility Barriers Finally, access to MIMIC-III requires completion of human subjects research training and agreement to a data use license. Although these requirements are minimal, they may pose barriers to entry. Additionally, the raw format of the dataset requires substantial preprocessing and domain knowledge, potentially impeding reproducibility and accessibility for non-experts [70].

4. Methodology: Feature Extraction from Clinical Text

This chapter describes the framework for extracting clinically relevant features from unstructured MIMIC-III notes. Section 4.1 outlines the feature selection process. Section 4.2 introduces the rule-based method (M1). Section 4.3 presents the selected LLM, followed by the note chunking method in Section 4.4, used in both the LLM-based method (M2) and the hybrid approach (M3). Section 4.5 details M2, while Section 4.6 describes M3, which combines rule-based precision with LLM-based contextual understanding. Experimental details are provided in Appendix D.

4.1. Features

The features used in this study are derived primarily from three established clinical scoring systems: the SOFA (Sequential Organ Failure Assessment) score (Table A.1 on page i), the SIRS (Systemic Inflammatory Response Syndrome) criteria (Table A.2 on page ii), and the quick SOFA (qSOFA) score (Table A.3 on page ii). These systems are widely utilized for assessing the severity of sepsis and guiding early intervention. All features are listed in Table A.4 on page iii.

Each feature was mapped to standardized medical ontologies to ensure semantic consistency and facilitate robust information retrieval. Specifically, features were cross-referenced with the Unified Medical Language System (UMLS) to obtain Concept Unique Identifiers (CUIs), which are later used for concept linking and disambiguation (see Section 4.2). Additionally, features were matched against the OMOP Common Data Model (CDM) to identify relevant synonyms and lexical variants, enhancing recall during text-based feature extraction. Supplementary mappings were performed with RxNorm, LOINC, and SNOMED CT to further enrich the feature vocabulary.

4.2. M1: Rule-Based Approach

The rule-based approach combines regular expression matching with biomedical concept recognition to extract symptom mentions from clinical notes. As illustrated in Figure 4.1, raw patient notes are first processed through regex-based detection of predefined clinical features. If a match is found, negation detection is applied. Otherwise, unmatched text is passed to a UMLS concept linker for secondary recognition. Both extracted symptoms and linked concepts are aggregated into a patient-level

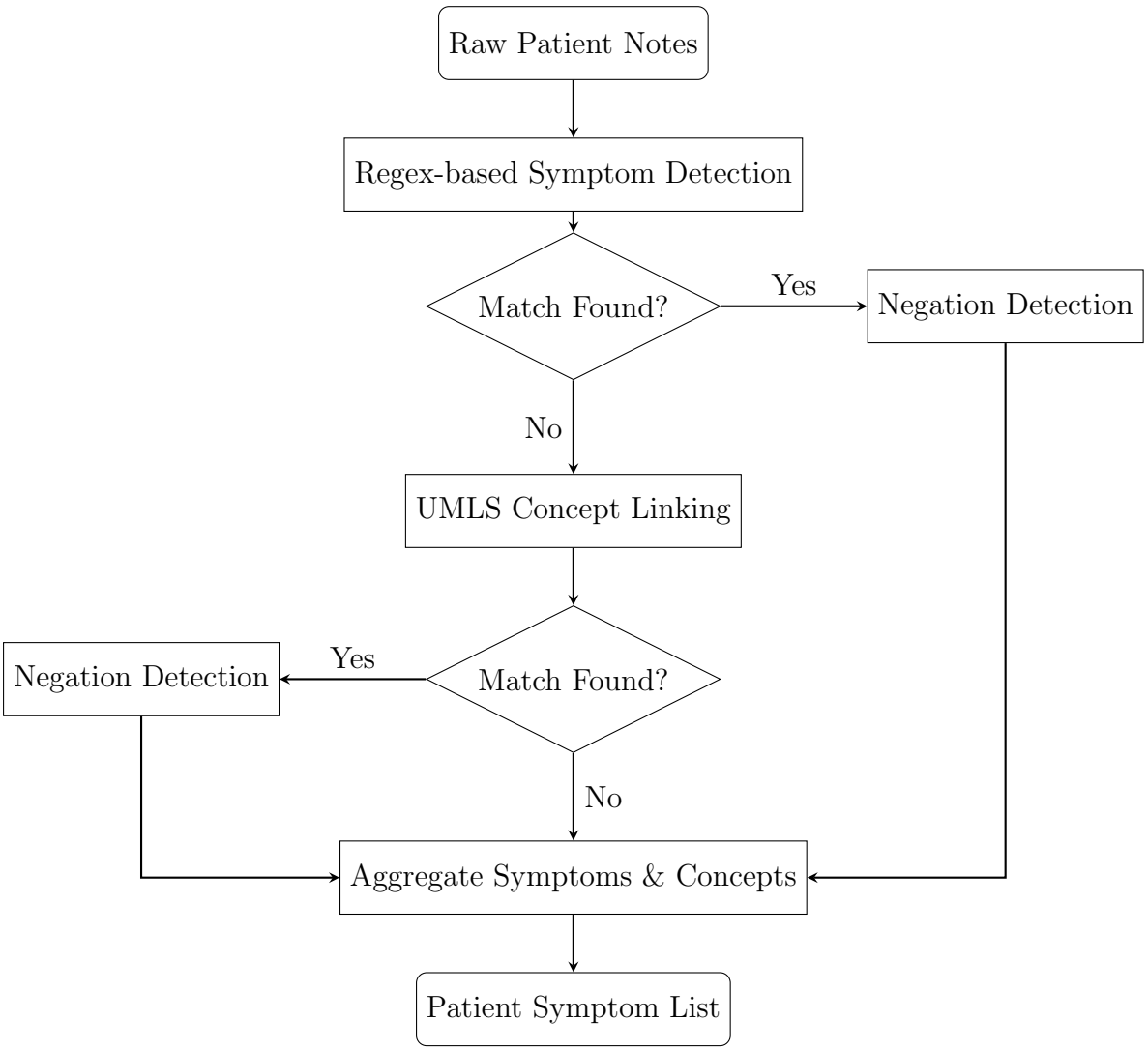

Figure 4.1.: This is an overview of Method 1: the *Rule-Based Approach*. Raw patient notes are processed using regular expressions to detect symptom mentions, including negations. If no match is found, the unmatched text is passed to a UMLS linker to identify relevant medical concepts. Both extracted symptoms and linked concepts are aggregated into a symptom list for each patient.

symptom list. This dual-path workflow ensures that features are captured even when lexical variability or abbreviation limits direct pattern matching.

Specifically, the rule-based component of the extraction pipeline targets predefined clinical features using two complementary methods: (i) handcrafted regular expressions tailored to common symptom expressions, and (ii) biomedical named entity recognition (NER) using the ScispaCy toolkit **scispacy**.

Regular Expressions For each predefined clinical feature (e.g., fever, tachypnea, creatinine level), regular expressions were developed through a multi-step process (Table A.5 on page v). First, OMOP vocabulary entries were reviewed to compile canonical terms and synonyms. Next, keyword searches across the corpus identified common linguistic variants and surface forms. Finally, regular expressions were manually crafted and iteratively refined to minimize false positives and negatives. Each expression was designed to be permissive enough to capture informal clinical language (e.g., "temp") while maintaining specificity.

Entity Recognition In parallel, the ScispaCy library was used to detect biomedical entities using the `en_core_sci_md` model, which offers a favorable trade-off between vocabulary coverage and runtime efficiency [35] . Detected entities were linked to UMLS Concept Unique Identifiers (CUIs), and only those matching the curated list of sepsis-related concepts were retained.

Feature Detection Logic A feature was considered present in a note chunk if either the regular expression or the ScispaCy model flagged a corresponding match. No comparative analysis or ensemble fusion was applied between the two methods; rather, detection was treated as a logical OR operation. Each match was annotated with metadata including the detected term, method of detection (regex or NER), and note section to support downstream validation and interpretability. This dual-path approach ensured that feature extraction was both flexible and comprehensive.

Evaluation M1 is included primarily as a baseline rather than a novel methodological contribution. Rule-based extraction with predefined vocabularies and regular expressions has been extensively explored in prior work [30], and the goal here is to use it as a stable point of comparison rather than to exhaustively evaluate it in isolation. Because the expressions were derived from established vocabularies that capture known lexical variation, I focused evaluation effort on the proposed LLM-based methods, where uncertainty and potential impact are greater.

Rather than relying on a manually annotated gold standard, M1 is assessed using a set of intrinsic diagnostics that provide indirect evidence of extraction behavior and utility:

- **Coverage proxy** The percentage of clinical notes containing at least one extracted symptom is reported to check whether the system extracts information from most notes. Although not a formal metric, low coverage may indicate that the system is missing relevant content or that the extraction rules are too narrow.

- **Top-N frequency distribution** The most frequently extracted symptoms are examined to see whether they align with typical clinical expectations for sepsis, providing a rough check on content relevance.

- **Regex match diversity** The number of different surface forms matched for each symptom is counted. A higher number suggests that the regular expressions capture a wide range of language variations and are effectively applied.

4.3. Large Language Models

Large language models (LLMs) have become essential tools in natural language processing (NLP) due to their generalization capabilities across a wide range of tasks. In this thesis, a LLM is used to extract features from clinical texts. Among available models, FLAN-XXL was chosen for its strong instruction-following abilities and its availability for local deployment, which is critical for working with sensitive medical data.

Security LLMs such as ChatGPT are typically accessed via APIs, which presents challenges when working with sensitive or credentialed datasets like MIMIC-III [77]. As hosted LLMs potentially transmit sensitive data to remote servers [78], [79], it raises substantial legal and ethical concerns especially within the European Union (EU), where the General Data Protection Regulation (GDPR) imposes strict limitations on the cross-border transfer of personal health data [79]. To mitigate these risks, this work relies exclusively on locally executed models, avoiding external data transfer during inference.

FLAN-XXL FLAN-XXL, a model developed by Google as part of the FLAN (Fine-tuned LAnguage Net) series, was selected due to its suitability for local deployment, which is essential when processing sensitive clinical data. With approximately 137 billion parameters, it is among the largest publicly disclosed transformer-based language models [80]. The choice of FLAN-XXL is further motivated by prior work from McInerney et al. (2023) [81], who demonstrated its effectiveness for zero-shot feature extraction from clinical notes. Their results indicate that the model's instruction-tuned architecture enables robust extraction of clinically relevant information without task-specific fine-tuning, closely aligning with the objectives of this thesis. Consequently, FLAN-XXL is employed in methods M2 and M3 (Sections 4.5 and 4.6).

4.4. Preprocessing Notes

Clinical notes were first deduplicated at the patient level by comparing the `raw note_text` field, ensuring that repeated content such as copied forward entries did not skew the downstream analysis. Following this, notes were temporally aligned using the patient's ICU admission (`intime`) and either the `sepsis_onset time` (for cases) or `control_onset_time` (for controls). Only notes documented between

admission and onset were retained and grouped into hourly bins relative to the onset event.

Each note was then segmented into semantically coherent chunks using a custom sentence-based splitting algorithm. Chunk boundaries were determined using punctuation and newline markers, with a strong preference for sentence-aligned splits. Chunks were constrained to contain a minimum of 256 characters and a maximum of 512 characters, ensuring both semantic richness and input-length efficiency. A 5-word overlap was introduced between adjacent chunks to preserve context across segment boundaries.

To ensure uniqueness and reduce embedding redundancy, identical chunks were further deduplicated within each patient-hour bin. The result was a structured, temporally binned, and context-preserving representation of clinical narratives suitable for retrieval-augmented inference models. This preprocessing step yielded 11,375 chunks for the case cohort and 92,954 chunks for the control cohort.

4.5. M2: Large Language Model Probabilistic Symptom Classification

The *Large Language Model Probabilistic Symptom Classification* method applies LLMs for probabilistic symptom classification based on unstructured clinical text. As illustrated in Figure 4.2, raw patient notes are divided into uniform chunks, which are embedded and stored in a vector database. For each symptom, relevant text segments are retrieved and combined with a predefined prompt before being processed by the LLM (FLAN-XXL) to estimate the probability of symptom presence. This workflow integrates chunk-based preprocessing, retrieval-augmented embedding, and symptom-specific analysis to enhance interpretability and clinical relevance in information extraction.

Retrieval-Augmented Generation Pipeline To structure the unstructured text data, clinical notes were aggregated on an hourly basis and represented as structured sequences of note chunks (see Section 4.4). These chunks were then processed using a Retrieval-Augmented Generation (RAG) pipeline, with feature-specific retrieval executed for each symptom or variable listed in Table A.4 on page iii. The pipeline employed the `hkunlp/instructor-large` model from HuggingFace's Instructor series to generate task-specific text embeddings. Each chunk was embedded using the prompt *"Represent the clinical text for retrieval:"* to guide the model toward learning retrieval-relevant representations. This instruction-based strategy has shown improved performance in medical retrieval tasks and aligns with prior methodology described by Shashikumar et al. (2025) [37].

Embeddings were stored in a Chroma vector database, chosen for its lightweight, efficient support for dense similarity search and structured metadata filtering [83]. Each embedded chunk was indexed along with metadata including `note_id` and

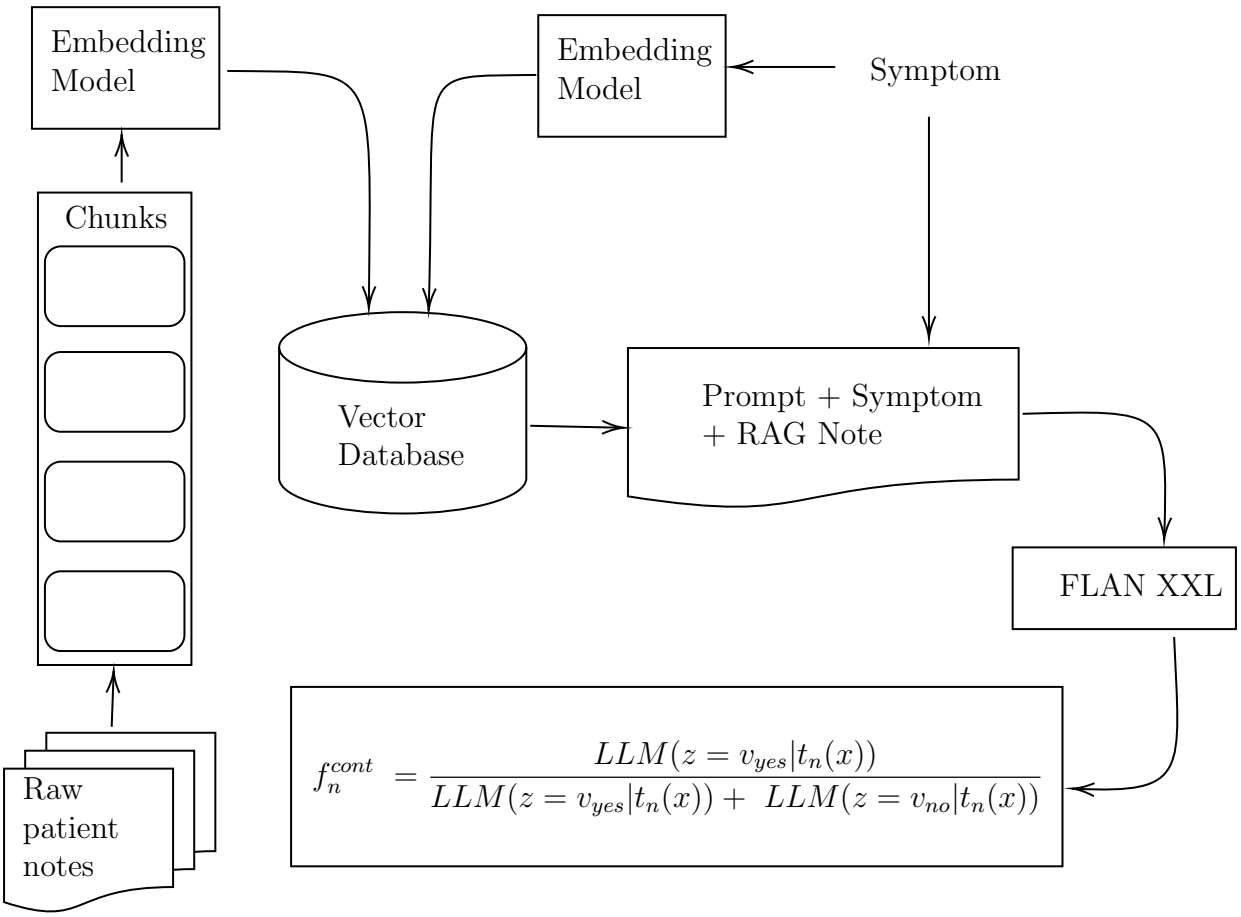

Figure 4.2.: This is an overview of Method 2: the *Large Language Model Probabilistic Symptom Classification*. For the function f_n^{cont}, $t_n(x)$ denotes the prompt generated by filling the template t_n with input x. The variable z refers to the token that immediately follows the prompt, and v_{yes} is the index of the token 'yes" in the model's vocabulary. This method draws inspiration from the work of McInerney et al. (2023) [39] and Zhang et al. (2023) [82].

`chunk_index`, enabling retrieval to be restricted to a specific patient's timeline.

For each target variable, a symptom query was embedded using the same instruction-tuned model. Retrieval was performed on a per-note basis, constrained to chunks sharing the same `note_id`, thereby preventing cross-patient data leakage. If a note contained three or fewer chunks, all were returned without querying the vector store. Otherwise, the top $K = 3$ most relevant chunks were retrieved based on cosine similarity. This fallback heuristic was introduced to preserve recall for short notes while maintaining semantic precision in longer documents. While prior work recommends larger retrieval windows (e.g., K=5) [37], these were not evaluated here. Because the retrieval pipeline is executed for every prediction rather than only in

ambiguous cases, larger K values would substantially increase computational cost and were therefore not considered.

Retrieved text segments were added as new columns in the patient-level dataframe, with one column per target variable, enabling further analysis and interpretability in downstream modeling stages. This modular design allows the embedded representations and retrieved content to be reused in later predictive tasks or integrated into generative clinical decision-support systems.

Prompts A prompting approach similar to McInerney et al. (2023) [39] is employed, utilizing the same prompt format. Given the scale of the dataset (553 unique icustay_ids and time bins for cases and 5,564 for controls, see Section 6.1) efficient processing is essential. Shorter prompts facilitate faster inference and lower memory consumption [84]. Each symptom is associated with a dedicated prompt and a corresponding question, as detailed in Table A.6 on page vi. The goal of the prompt is to receive a yes or no answer to the question whether the patient at that time reaches a certain threshold that awards a point in either the SOFA-score, qSOFA-score or SIRS-criteria. The questions ask whether it is at the point the note is being made as the time of the measurement is important for the prediction model.

Prompt for M2 based on McInerney et al. (2023) [39]

Read the following text from a clinical note:

< RAG_chunk >

< question >

Confidence Score To extract high-level features from clinical text, instruction-tuned LLMs are employed for zero-shot inference using a set of expert-defined prompt templates $t_1, \ldots, t_N$. This approach follows the method introduced by McInerney et al. (2023) [39], where each prompt elicits a binary response corresponding to a clinical symptom or feature.

Each binary feature is defined as:

$$f_n^{\text{binary}} = I\left[\arg\max_z \text{LLM}(z \mid t_n(x)) = v_{\text{yes}}\right] \tag{4.1}$$

where $t_n(x)$ represents the prompt instantiated with input x, z is the predicted token, and v_{yes} is the index of the token "yes" in the vocabulary.

To account for prediction uncertainty, the continuous probability associated with

the "yes" response is used instead of the binary indicator, as in:

$$f_n^{\text{cont}} = \frac{\text{LLM}(z = v_{\text{yes}} \,|\, t_n(x))}{\text{LLM}(z = v_{\text{yes}} \,|\, t_n(x)) + \text{LLM}(z = v_{\text{no}} \,|\, t_n(x))} \tag{4.2}$$

This continuous formulation allows for incorporating model confidence into downstream tasks. For each target variable, the maximum value across all note chunks within the same chart time was selected as the final input feature for downstream modeling.

Evaluation Similar to the previous methodology a manually annotated test set is absent, therefore system performance was evaluated using intrinsic metrics, allowing for indirect assessment of extraction quality.

- **Boxplots** Boxplots were generated for both case and control groups and subsequently analyzed to compare the distribution of values between the two cohorts.
- **Clinician review** A stratified sample of 10 model predictions per symptom was manually reviewed by the author. For each symptom, predictions were categorized as positive or negative based on a classification threshold of 0.5. The proportion of positive and negative examples selected per symptom reflected the overall class distribution, with a minimum of three positive and three negative instances included for each. Cases with uncertain clinical relevance or ambiguous model outputs were subsequently reviewed and validated in consultation with a clinical expert to ensure medical accuracy.

4.6. M3: Hybrid Rule-Guided Large Language Model Quantification

The *Hybrid Rule-Guided Large Language Model Quantification* method builds on the symbolic precision of M1 and the contextual adaptability of M2 by introducing a targeted information extraction framework. As depicted in Figure 4.3, raw clinical notes are first segmented into chunks, which are screened for symptom mentions using regular expressions and UMLS-based concept linking. Chunks containing target symptoms are then passed to the LLM (FLAN-XXL), which is prompted to extract numerical measurement values. Rather than classifying symptom presence, this approach retrieves exact physiological or laboratory values from relevant text segments.

UMLS Chunks Using the strategy described in Section 4.4, clinical notes are binned and segmented into chunks. These chunks are then processed through the same pipeline as M1, but with relaxed regular expression rules for symptoms associated with

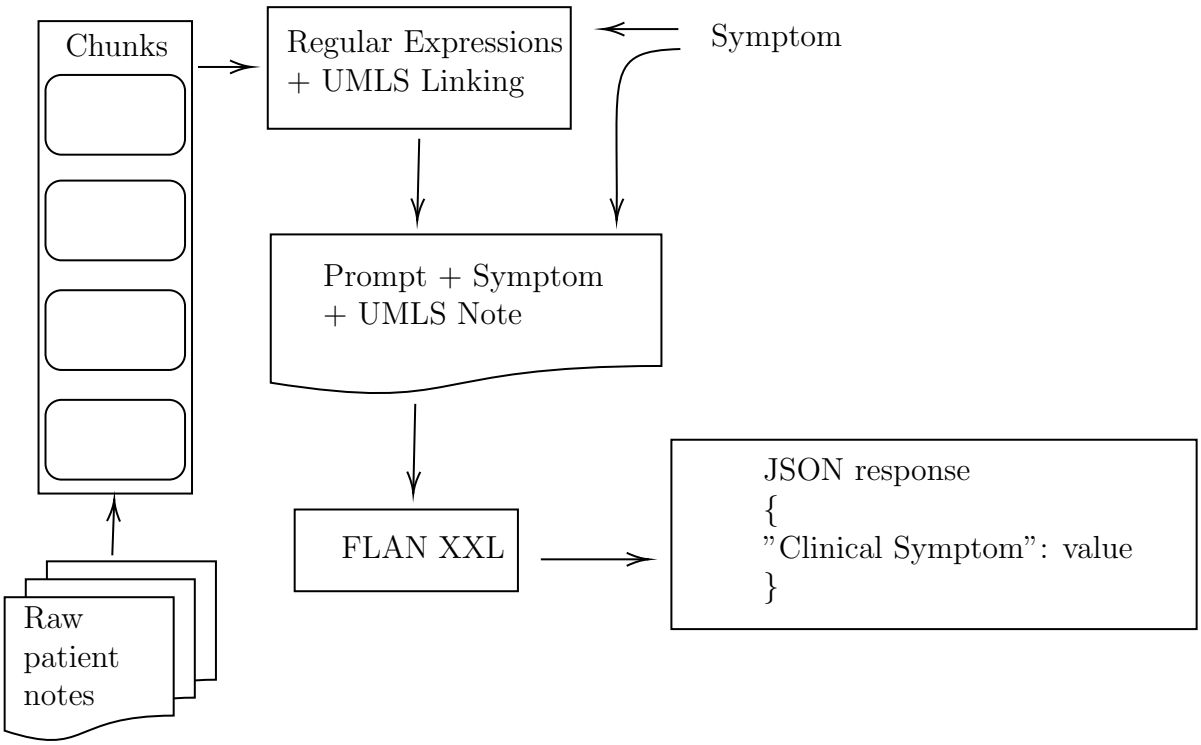

Figure 4.3.: This is an overview of Method 3: the *Hybrid Rule-Guided LLM Large Language Model Quantification*. Raw clinical notes are first segmented into manageable chunks. Each chunk is screened for symptom mentions using a combination of regular expressions and UMLS-based concept linking. Chunks identified as containing target symptoms are then passed to a LLM, which is prompted to extract associated measurement values.

numerical values. For instance, *fever* is generalized to *temperature*, and *leukocytosis* to *white blood count*, since such symptoms may appear either as named entities or as isolated measurements. Pre-selecting relevant chunks helps reduce hallucinations, which is essential since LLMs still generate too many factual errors for reliable clinical implementation [85].

Prompt Each extracted text chunk was inserted into a prompt designed to elicit exact numerical measurements from the language model. Inspired by the work of Bhattarai et al. (2024) [32] on knowledge-guided generation, a templated instruction format was constructed. The prompt included: (1) a measurement-specific question, (2) a strict instruction to return only exact values or null, (3) a fixed output schema in JSON format, and (4) one or more in-context examples demonstrating the expected behavior.

All prompt questions and corresponding in-context examples were derived from

Prompt for M3 based on Shashikumar et al. (2025) [37]

< question >
Only answer if you know the EXACT measurement values, else return null. Return your answer ONLY in this exact JSON format: "value": < number or null >

Here are some examples of text and the JSON output.
< examples >

Text: < UMLS_chunk >

a manually curated table (see Table A.7 on page vii and Table A.8 on page viii). Features that referred to the same underlying physiological measurement, such as "fever" and "hypothermia" both indicating body temperature, were consolidated under a unified variable label to ensure consistent downstream handling.

The model used for generation was `FLAN-T5-XXL`, loaded with mixed-precision settings on GPU resources to optimize memory and runtime performance. Each prompt was batched and fed into the model using HuggingFace's generate() API, and the outputs were decoded using the accompanying tokenizer. For each input chunk, only the first valid model output (i.e., one that did not return null) was retained.

To ensure adherence to the format, only responses starting with the `"value"` key were parsed. This requirement originated from the instruction prompt and acted as a structural constraint, ensuring compatibility with downstream parsing scripts. The raw model outputs were then passed to a separate postprocessing pipeline for numeric extraction, unit normalization, and validation.

Output and Postprocessing The outputs from the language model were postprocessed to extract structured numeric measurements. The model was explicitly instructed to return responses in JSON format, using the key `"value"` to indicate either a number or `null`. Accordingly, only predictions beginning with the token `"value": <number>` were retained for further parsing. Outputs where `"value"` was set to `null` were interpreted as missing and converted to `NaN` during postprocessing. This formatting requirement served as both a structural constraint and a safeguard against inadvertently extracting unrelated or noisy numerical content.

To normalize and validate values, a set of tailored parsing functions was applied. For each clinical variable, a list of acceptable unit variants was manually curated by inspecting the language model's outputs. Units that were deemed semantically equivalent, such as 'mcg/kg/min', 'mcg/kg', and 'mcg', were grouped together and accepted for the same variable. This approach accounted for variability in model output formatting without attempting semantic inference over units.

Temperature required special handling. Despite the prompting strategy exclusively

including Fahrenheit examples, the model occasionally returned temperatures in both Fahrenheit and Celsius within the same output. To ensure consistency and avoid mixed-unit data, only values within the physiologically plausible Fahrenheit range (90–110°F) were retained. Any additional values outside this range were discarded as they likely reflected the Celsius range.

White blood cell counts also presented unit variability. Some values were formatted with an explicit K/uL unit (denoting thousands of cells per microliter), while others appeared as raw numbers (e.g., 9000 instead of 9.0). Since the majority of outputs followed this convention and units were often omitted, the extraction logic was calibrated to interpret the values consistently as thousands per microliter. Variable-specific value ranges were applied as filters to exclude physiologically implausible values and improve data quality based on value ranges reported by Wang et al. (2020) [86].

Finally, any values that failed to meet structural, unit, or range criteria were set to NaN, serving as an explicit marker for invalid or ambiguous predictions. The resulting post-processed dataset was a structured DataFrame with cleaned numeric values per variable, suitable for downstream statistical analysis.

Evaluation The method was evaluated through targeted manual validation of extracted numerical measurements against the original clinical notes. In contrast to the evaluation of M2, this process did not require medical expertise, as extraction was limited to a predefined set of physiological and laboratory variables identified through regular expressions and UMLS-based concept linking, resulting in substantially lower semantic ambiguity. A prediction was considered correct if the extracted value matched the measurement described in the source note after unit normalization and postprocessing.

For each variable, ten UMLS-linked text chunks were sampled in proportion to the observed extraction frequency, with a minimum of three valid and three invalid or null extractions per variable to ensure coverage of heterogeneous documentation patterns and common failure modes. This evaluation was designed to verify numeric extraction correctness under constrained conditions, rather than to provide a comprehensive estimate of recall across all note types.

Motivation and Differentiation In contrast to M1, which fails on non-canonical phrasing and M2, which only classifies presence without magnitude, M3 retrieves quantitative and interpretable features directly. This capability is crucial for downstream modeling tasks that depend on numeric trends or thresholds.

5. Evaluation: Feature Extraction from Clinical Text

This section evaluates the performance of three feature extraction methods (rule-based (M1), LLM-based classification (M2), and a hybrid rule-guided LLM approach (M3)) on the task of extracting sepsis-relevant features from unstructured clinical text. The evaluation is structured around the following overarching research question:

> **RQ 1:** How accurately can regular expressions, LLMs, and hybrid models extract sepsis-related clinical features from unstructured clinical notes?

This overarching question is addressed through the following subquestions, each targeting a specific method and evaluating aspects such as coverage, consistency, clinical validity, and interpretability:

1.1 How accurately does the rule-based approach with regular expressions and UMLS-linking (M1) extract sepsis-related clinical features from unstructured clinical notes? (Section 5.1)

1.2 How accurately does the zero-shot LLM approach with probabilistic symptom classification (M2) extract sepsis-related clinical features from unstructured clinical notes? (Section 5.2)

1.3 How accurately does the hybrid rule-guided few-shot LLM method (M3) extract sepsis-related clinical features from unstructured clinical notes? (Section 5.3)

1.4 How do the three approaches (M1, M2, and M3) compare in terms of coverage, granularity, and interpretability? (Section 5.4)

5.1. M1: Rule-Based Approach

The following research question guides the evaluation of the rule-based approach:

1.1 How accurately does the rule-based approach with regular expressions and UMLS-linking (M1) extract sepsis-related clinical features from unstructured clinical notes? This overarching question is addressed through the following subquestions:

   a. To what extent do rule-based methods consistently detect key sepsis-related features across clinical notes?

b. How robust are the regular expressions to linguistic variability in clinical language?

c. Which clinical indicators are systematically missed due to the limitations of fixed pattern matching?

These questions guide the analysis of M1. The evaluation focuses on feature-level coverage and qualitative error patterns, examining instances of correct detection, missed extractions, and failures attributable to linguistic variability in clinical documentation. Particular attention is paid to differences in extraction behaviour across feature types, enabling a systematic assessment of the strengths and limitations of fixed pattern matching in the clinical setting. These aspects are examined below in alignment with subquestions (a)–(c).

1.1 a. To what extent do rule-based methods consistently detect key sepsis-related features across clinical notes?

A set of intrinsic metrics, focusing on symptom coverage and extraction reliability, have been employed to evaluate the consistency of the rule-based extraction approach.

Coverage Proxy (Heuristic Evaluation): Overall coverage was approximated by calculating the proportion of clinical notes in which at least one sepsis-related symptom was identified. The results are as follows:

- **Control Group:** Symptoms were detected in 84.02% of clinical notes.
- **Case Group:** Symptoms were detected in 91.58% of clinical notes.

These results indicate that the rule-based method was able to detect at least one sepsis-related symptom in the majority of clinical notes for both groups. As expected, the detection rate was higher in the case group (91.58%) compared to the control group (84.02%), reflecting the greater prevalence of sepsis symptoms among confirmed cases. This difference supports the face validity of the approach, suggesting that the method is sensitive to clinically meaningful variation in symptom presentation between cohorts.

Top-N Symptom Frequency Distribution: Table 5.1 presents the most frequently detected sepsis-related features in both case and control groups, expressed as both absolute counts and the percentage of clinical notes in which each symptom was identified.

In the case group, *artificial respiration* (67.46%), *creatinine* (53.01%), *tachycardia* (37.32%), and *fever* (34.13%) were the most prevalent features. These findings are consistent with the expected clinical presentation of sepsis, which often involves respiratory support, impaired renal function, elevated heart rate, and fever.

Similarly, in the control group, the top symptoms included *artificial respiration* (57.45%), *creatinine* (40.88%), *tachycardia* (26.53%), and *fever* (26.48%). While

Symptom	Case Count	Case %	Control Count	Control %
artificial respiration	593	67.46%	5044	57.45%
creatinine	466	53.01%	3589	40.88%
tachycardia	328	37.32%	2329	26.53%
fever	300	34.13%	2325	26.48%
systolic blood pressures	206	23.44%	1701	19.37%
urine output	189	21.50%	1136	12.94%
leukocytosis	182	20.71%	1063	12.11%
partial pressure of oxygen in arterial blood	173	19.68%	709	8.08%
mean arterial pressure	172	19.57%	353	4.02%
norepinephrine	141	16.04%	66	0.75%
tachypnea	117	13.31%	650	7.40%
hypothermia	110	12.51%	597	6.80%
dopamine	104	11.83%	166	1.89%
blood platelets	74	8.42%	413	4.70%
epinephrine	71	8.08%	253	2.88%
bilirubin	28	3.19%	204	2.32%
leukopenia	27	3.07%	206	2.35%
dobutamine	19	2.16%	23	0.26%
glasgow coma score	11	1.25%	104	1.18%

Table 5.1.: Top Detected Symptoms and Their Prevalence Across Clinical Notes.

these features also appeared frequently in controls, their relative proportions were consistently lower compared to the case group, reflecting the expected higher prevalence of critical symptoms among confirmed sepsis cases.

Notably, pharmacologic interventions such as *norepinephrine* (16.04% vs. 0.75%) and *dopamine* (11.83% vs. 1.89%) appeared far more frequently in the case group, likely indicating the use of vasopressor support in response to septic shock [87]. These differences underscore the method's capacity to differentiate between cohorts based on symptom burden and treatment intensity.

Overall, the percentage-based frequency analysis strengthens the face validity of

the rule-based system, demonstrating that it effectively captures clinically salient features more commonly associated with sepsis.

1.1 b. How robust are the regular expressions to linguistic variability in clinical language?

Linguistic Robustness Robustness to linguistic variability in clinical language was assessed by analyzing the diversity of surface forms matched for each symptom. This approach reflects how well each pattern generalized across different lexical and syntactic realizations.

Results showed substantial variability in regex flexibility across symptoms. Some concepts such as *fever* (69 unique forms, e.g., "febrile", "pyrexia", "100.4 f") and *leukocytosis* (145 forms, e.g., "wbc 14.3", "leukocytosis") exhibited high lexical diversity, indicating that their regex patterns captured a wide range of expressions. Other symptoms, such as *dobutamine*, *glasgow coma score*, and *mean arterial pressure*, each matched only a single canonical form (e.g., "dobutamine", "gcs", "map"), reflecting minimal surface variation.

Overall, the feature-wise diversity of surface forms (ranging from 1 to 145 unique matches) highlights the importance of tailoring regex design to the linguistic variability inherent in each clinical concept. Features with high variation (e.g., *fever*, *hypothermia*, *tachycardia*) benefited from flexible patterning, while rigid expressions were more prone to undercoverage.

1.1 c. What important clinical indicators are systematically missed due to fixed pattern limitations?

Fixed-Pattern Limitations and Systematic Misses A manual analysis of the M1 extraction outputs was conducted to identify clinical indicators systematically missed due to fixed pattern limitations (see Sections 5.2 and 5.3). These limitations largely reflect the inability of regular expressions to capture implicit expressions, lexical variability, and contextual nuance in clinical language.

Indicators expressed through non-standard abbreviations, rare synonyms, or implicit phrasing were frequently missed. For example, the abbreviation "levophed" for *norepinephrine* or "oliguria" for *urine output* was not captured by existing patterns. In contrast, concepts like *creatinine* (e.g., "Cr") were consistently detected due to their predictable and standardized representation.

Another critical limitation is that M1 does not extract measurement values. Instead, it only detects whether a concept is mentioned, without interpreting whether it is clinically significant. A few exceptions exist (*fever*, *hypothermia*, *tachycardia*, *tachypnea*, *leukocytosis*, and *leukopenia*) for which regular expressions include numeric thresholds (e.g., temperatures above 100°F to identify fever). However, even these are limited in scope; for instance, "temp 101" or "T=101.2" will be overlooked if they do not match the exact regex pattern.

As a result, this method does not account for contextual information or measurement values, leading to an overrepresentation of some entities. At the same time, many clinically relevant instances go undetected unless there is an exact surface form match, resulting in an underrepresentation of symptoms in downstream analyses. These findings suggest that supplementing rule-based extraction with machine learning or hybrid approaches could improve recall, particularly for symptoms with high linguistic variability or those defined by measurements.

5.2. M2: Large Language Model Probabilistic Symptom Classification

The following research question guides the evaluation of the LLM approach:

1.2 How accurately does the zero-shot LLM approach with probabilistic symptom classification (M2) extract sepsis-related clinical features from unstructured clinical notes? This overarching question is addressed through the following subquestions:

a. How accurately does the zero-shot LLM approach with probabilistic symptom classification (M2) extract sepsis-related clinical features from unstructured notes?

b. Which clinical features are reliably extracted, and where does M2 encounter limitations?

c. What are the key strengths and limitations of M2 in handling ambiguity, implicit context, and fine-grained clinical features?

These questions guide the analysis of M2. The evaluation focuses on the model's handling of semantic variability, contextual dependence, and implicit clinical information, with particular attention to synonym ambiguity, context-dependent variables such as the Glasgow Coma Scale (GCS), implicit inference cases (e.g., creatinine in end-stage renal disease), and sensitivity to numerical thresholds. These aspects are examined below in alignment with subquestions (a)–(c).

1.2 a. How accurately does the zero-shot LLM approach with probabilistic symptom classification (M2) extract sepsis-related clinical features from unstructured notes?

Clinician Review: Overview Results A total of 190 clinical notes were annotated, including 46 that received additional review by a clinical expert. Overall, the M2 model achieved an accuracy of 0.80, recall of 0.95, precision of 0.61, and an F1-score of 0.74 (Table 5.2). These results highlight strong sensitivity with relatively fewer false negatives, balanced by moderate precision.

Feature	TP	FP	TN	FN	Acc.	Rec.	Prec.	F1-score
dobutamine	1	3	6	0	0.70	1.00	0.25	0.40
dopamine	3	2	5	0	0.80	1.00	0.60	0.75
epinephrine	4	0	6	0	1.00	1.00	1.00	1.00
norepinephrine	1	4	3	2	0.40	0.33	0.20	0.25
glasgow coma score	2	1	7	0	0.90	1.00	0.67	0.80
bilirubin	3	2	4	1	0.70	0.75	0.60	0.67
creatinine	6	0	4	0	1.00	1.00	1.00	1.00
partial pressure of oxygen in arterial blood	2	2	6	0	0.80	1.00	0.50	0.67
blood platelets	2	3	5	0	0.70	1.00	0.40	0.57
urine output	3	4	3	0	0.60	1.00	0.43	0.60
artificial respiration	3	2	5	0	0.80	1.00	0.60	0.75
mean arterial pressure	3	2	5	0	0.80	1.00	0.60	0.75
fever	5	0	5	0	1.00	1.00	1.00	1.00
hypothermia	3	2	5	0	0.80	1.00	0.60	0.75
tachycardia	5	0	5	0	1.00	1.00	1.00	1.00
tachypnea	2	3	5	0	0.70	1.00	0.40	0.57
leukocytosis	3	1	6	0	0.90	1.00	0.75	0.86
leukopenia	1	3	6	0	0.70	1.00	0.25	0.40
systolic blood pressures	3	1	6	0	0.90	1.00	0.75	0.86
all symptoms	55	35	97	3	0.80	0.95	0.61	0.74

Table 5.2.: Classification Performance Metrics for each M2 Feature.

Clinician Review: Low False Negatives Reflect Strong Sensitivity Prompted large language models were used to classify whether symptoms were present or absent in context. The model demonstrated strong contextual understanding, correctly identifying symptoms such as altered mental status, hypotension, and respiratory distress even when no explicit measurement was provided. For example, the phrase "Physical Examination General: Not responsive." was accurately interpreted as indicating a *Glasgow Coma Score* below 14, while "hct dropped to 20, hypotensive

and tachycardic" was correctly classified as positive for a *Mean Arterial Pressure* below 70 mmHg.

Clinician validation confirmed the model's high sensitivity. Out of 190 reviewed predictions, there were 55 true positives (TP) and only 3 false negative (FN), yielding a recall of 0.95. This extremely low FN rate suggests that the model reliably identifies relevant clinical symptoms when present, even in the absence of explicit markers, highlighting its strong contextual reasoning abilities.

Clinician Review: False Positives Indicate Overgeneralization While the model rarely missed relevant symptoms, it more frequently over-predicted their presence. Clinician review revealed 35 false positives (FP) compared to 97 true negatives (TN), with a precision of 0.61. These FPs often arose from ambiguous, outdated, or loosely associated mentions. For instance, symptoms inferred from the past, such as medications that were "on hold", or from indirect evidence, such as equating nasal cannula oxygen use ("NC 2lit/min started") with vasopressor support, led to incorrect symptom attribution. These cases indicate that while the model can reason effectively, it also tends to overgeneralize or overestimate symptom presence in borderline or contextually unclear scenarios.

Confidence Score Trends Across Clinical Features Confidence score distributions across key symptom categories are shown in Figures B.1–B.3 in Appendix B on page xi. These plots compare the model's predicted probabilities for case and control groups, offering insight into how feature-specific patterns align with clinical expectations. Most notably, the median of cases for each feature is consistently higher than the control group.

1.2 b. Which clinical features are reliably extracted, and where does M2 encounter limitations?

High-Performing Features Symptoms such as *epinephrine*, *creatinine*, *fever*, and *tachycardia* were perfectly detected, each achieving an F1-score of 1.00 with high precision and recall. Interestingly, both *fever* and *tachycardia* are defined by threshold values but can also be identified through textual cues (e.g., "febrile," "tachycardic"), meaning the model succeeded both by reading measurements and by recognizing descriptive language. Other variables, including *leukocytosis* and *systolic blood pressure*, also performed strongly (F1 $\geq$ 0.85). By contrast, features such as *dobutamine* were less reliably extracted, with low precision and F1-scores (around 0.40) despite high recall.

Context-Dependent Features Some features benefited directly from M2's ability to interpret indirect information and contextual cues. For example, the *Glasgow Coma Score* was often inferred from descriptive statements such as "not responsive" or "does not move any extremities," even when no specific numerical score was documented.

These cases illustrate how M2 can extract clinically meaningful information beyond explicit mentions, providing broader coverage than rule-based approaches.

While *Glasgow Coma Score* was the clearest example of this strength, other features also benefited from contextual reasoning. Clinician feedback highlighted an instance where the model correctly inferred abnormal *creatinine* levels despite the absence of explicit measurements:

> The clinical note does not contain any information about the patient's creatinine level. BUT: "ESRD management with HD per renal team."
>
> ESRD = *end-stage renal disease* (terminal kidney failure). HD = *hemodialysis*. Patients with end-stage renal failure require dialysis because their kidneys no longer function adequately, which typically results in creatinine levels well above 1.2,mg/dL.

This example underscores how M2 can capture implicit clinical meaning that would be inaccessible to rule-based extraction, though such inferences also raise questions about overgeneralization and annotation consistency.

Lexical Errors Other features were falsely extracted due to lexical confusions. Norepinephrine, for example, had the lowest precision (0.20) and the highest number of false positives (4), partly because the model incorrectly classified mentions of the abbreviation "neo" as evidence of norepinephrine administration. Similarly, mentions of "dopamine 2.5 mcg/kg/min" sometimes triggered positive predictions for *dobutamine*, suggesting difficulty distinguishing between lexically or semantically similar features.

Numerical Errors Urine output was another weak point. With four false positives and a precision of only 0.40, the model often misclassified cases because urine output is usually documented hourly, whereas the SOFA score requires a 24-hour aggregate. Nonetheless, M2 occasionally succeeded by leveraging clinical context. As the clinical annotator noted:

> "No specific numerical values, but since oliguria is defined as less than 500 ml in 24 hours, the criterion should be met."

In this instance, the model correctly inferred that the SOFA definition applied.

Numerical errors also affected *leukopenia*, which showed very low precision (0.25). For example, the model misclassified a "WBC of 33" as leukopenia because it interpreted the number as "<4.0E9/L," even though in clinical notation "33" corresponds to 33E9/L and is therefore within the normal range. The only instance where the model correctly identified *leukopenia* was when the note explicitly stated, "Hematology: Leukopenia of unknown etiology." These examples highlight systematic weaknesses in the model's numerical reasoning, as it struggles when values are implicit or depend on unit conventions.

Future Work Prompt engineering offers a promising direction for reducing numerical errors, for example by clarifying expected units or explicitly framing cutoff thresholds. Structured post-processing (e.g., normalizing lab values and enforcing unit checks) could further mitigate these systematic mismatches.

Summary In summary, M2 performed reliably for explicitly stated symptoms (e.g., *fever*, *tachycardia*). At the same time, errors frequently stemmed from synonym confusions (e.g., "neo" for norepinephrine), unit mismatches (e.g., "WBC of 33" for *leukopenia*), and temporal inconsistencies (e.g., hourly urine output versus SOFA's 24-hour criterion). Beyond explicit mentions, M2 also showed strength in contextual reasoning: features such as *Glasgow Coma Score* were successfully inferred from descriptive or indirect cues, extending coverage beyond rule-based methods.

1.2 c. What are the strengths and limitations of M2 in handling ambiguity and implicit context?

Strengths in Contextual Reasoning One of the key strengths of M2 lies in its ability to interpret implicit information. The model successfully inferred clinical meaning even when no explicit measurement was provided. For example, phrases such as "not responsive" were correctly interpreted as indicating a *Glasgow Coma Score* below 14, despite the absence of numerical values.

> "Somewhat difficult: The clinical notes do not contain any explicit information about the Glasgow Coma Scale score. However, the description of the neurological condition ("Neuro exam remains poor," "does not move any extremities," "no gag or cough") indicates a very low score. A specific numerical score is not mentioned in the text."

Beyond *Glasgow Coma Score*, clinician feedback highlighted how M2 captured implicit context for other features. For example, it correctly inferred abnormal *creatinine* levels from the phrase "ESRD management with HD per renal team," recognizing that patients on dialysis almost always present with creatinine well above 1.2,mg/dL.

Similarly, bilirubin levels could sometimes be inferred from qualitative descriptions, such as "mild hyperbilirubinemia," which implies a value above the diagnostic threshold.

> "Although no explicit numerical value is given, the term "mild hyperbilirubinemia" would imply a value of more than 1.2 milligrams per deciliter."

These examples illustrate how M2 moves beyond surface-level keywords and extracts clinically meaningful signals from implicit cues, outperforming rule-based methods in flexibility.

Annotation Challenges At the same time, features like *Glasgow Coma Score* expose the limitations of contextual inference and annotation reliability. Contradictory statements within the same note often prevented a definitive label:

> "The clinical notes do not contain any explicit information about the Glasgow Coma Scale score. The information about the patient's condition is contradictory: One section on the physical examination describes the patient as "not responsive," which would correspond to a very low score. But a later entry states that the patient is "currently interactive," which would indicate a higher GCS score. Therefore, no definitive statement about the current score can be made based on the available text. But if absolutely necessary, then probably under 14."

Larger context windows could help resolve contradictory information for features like *Glasgow Coma Scale*, potentially containing more specific information about when contradicting information has taken place.

Similarly, bilirubin annotations proved challenging. While "mild hyperbilirubinemia" implies a value above 1.2,mg/dL, notes with phrases like "mildly elevated LFTs" were less clear:

> "Although no explicit numerical value is listed, the clinical note mentions that the liver function tests (LFTs) are 'mildly elevated'. However, an increase in LFTs does not necessarily indicate a bilirubin level above 1.2 mg/dL, but in combination with ascites it is quite possible."

These quotations highlight both the potential and the limits of the model's contextual inference. At the same time, it underscores a broader challenge for annotation: when the text itself is ambiguous or lacks explicit values, even expert annotators may struggle to determine whether bilirubin levels are above 1.2 mg/dL. Such cases illustrate that performance metrics partly reflect not only model limitations but also the inherent uncertainty of the underlying clinical documentation. However, annotation protocols could be refined to better handle ambiguous cases, ensuring consistency across clinical reviewers. Finally, expanding the annotated dataset would improve the statistical robustness of feature-level evaluations.

Overgeneralisation The same contextual reasoning that allows M2 to recover implicit meaning can also lead to overgeneralisation. Ambiguous cues were sometimes interpreted as strong evidence for unrelated features, producing false positives. For example, oxygen therapy ("NC 2lit/min started") was misclassified as vasopressor support. These errors suggest that while M2 is capable of leveraging indirect context, it can also overextend its inferences, blurring the line between genuine clinical reasoning and a systematic bias toward positive classification.

Summary Overall, M2 demonstrates clear strengths in semantic and contextual reasoning, capturing implicit clinical meaning that rule-based methods would miss. However, it struggles with numerical thresholds, temporal aggregation, overgeneralization, and contradictions in documentation. These findings underscore both the promise and the limits of large language models in clinical feature extraction: they excel at interpreting ambiguous or implicit language, but reliable deployment will require advances in context management, prompt design, and annotation methodology.

5.3. M3: Hybrid Rule-Guided Large Language Model Quantification

The following research question guides the evaluation of the LLM approach:

1.3 How accurately does the hybrid rule-guided few-shot LLM method (M3) extract sepsis-related clinical features from unstructured clinical notes? This overarching question is addressed through the following subquestions:

a. To what extent does the hybrid method capture feature presence and values that align with clinical interpretation?

b. What types of clinical variables are not reliably extracted by the hybrid method?

c. Which types of clinical variables are best suited for hybrid extraction?

These questions guide the analysis of M3. The evaluation focuses on the correctness and reliability of extracted numerical values, examining how well the hybrid method aligns extracted measurements with clinical interpretation. Particular attention is paid to differences across variable types, including standardized laboratory values, vital signs, and more context-dependent or ambiguously documented measurements. This analysis enables a systematic assessment of which clinical variables are well suited to hybrid rule-guided quantification and where limitations remain. The results are discussed below in alignment with subquestions (a)–(c).

1.3 a. To what extent does the hybrid method capture feature presence and values that align with clinical interpretation?

Feature Extraction Performance The hybrid method in M3 demonstrates strong alignment between extracted features and clinical interpretation, as evidenced by high F1-scores across a majority of variables. Features such as *epinephrine* (F1 = 0.92), *norepinephrine* (F1 = 1.00), *bilirubin* (F1 = 1.00), and *creatinine* (F1 = 0.93) highlight the model's ability to reliably capture both the presence and correct value of key physiological measurements. High performance was also observed for *glasgow coma score* (F1 = 0.94), *white blood count* (F1 = 0.94), and *blood platelets*

Feature	TP	FP	TN	FN	Acc.	Rec.	Prec.	F1-score
dobutamine	3	0	7	1	1.00	1.00	1.00	1.00
dopamine	3	0	7	1	1.00	1.00	1.00	1.00
epinephrine	6	1	3	0	0.90	1.00	0.86	0.92
norepinephrine	7	0	3	0	1.00	1.00	1.00	1.00
glasgow coma score	8	1	1	0	0.90	1.00	0.89	0.94
bilirubin	5	0	5	0	1.00	1.00	1.00	1.00
creatinine	7	1	2	0	0.90	1.00	0.88	0.93
partial pressure of oxygen in arterial blood	8	0	0	2	0.80	0.80	1.00	0.89
blood platelets	5	1	4	0	0.90	1.00	0.83	0.91
urine output	0	0	10	0	1.00	NaN	NaN	NaN
mean arterial pressure	3	3	4	0	0.70	1.00	0.50	0.67
temperature	6	1	2	1	0.80	0.86	0.86	0.86
heart rate	5	2	2	1	0.70	0.83	0.71	0.77
respiratory rate	2	1	7	0	0.90	1.00	0.67	0.80
white blood count	8	0	1	1	0.90	0.89	1.00	0.94
systolic blood pressure	1	2	7	0	0.80	1.00	0.33	0.50
all symptoms	77	13	65	5	0.89	0.94	0.86	0.90

Table 5.3.: Classification Performance Metrics for each M3 Feature

(F1 = 0.91), indicating that the model's outputs reflect clinically meaningful and contextually grounded information.

These results are not solely attributable to model architecture. The elevated F1-scores stem in part from the incorporation of UMLS-linked chunk selection, which ensured that the model was only prompted with relevant and semantically coherent input text. This pre-selection strategy greatly reduced the likelihood of spurious generations. Additionally, a rigid post-processing pipeline enforced strict output formats and applied variable-specific validation rules, further minimizing hallucinations which is an issue known to affect large language models in clinical settings [85]. However, the rigid post-processing steps do lead a higher precision while the sensitivity cannot be guaranteed.

While aggregate metrics for *all symptoms* show high recall (0.94) and F1-score

Feature	Count	Feature	Count
heart rate	2914	respiratory rate	170
fever	2905	bilirubin	79
creatinine	2267	norepinephrine	76
white blood count	2052	glasgow coma score	75
partial pressure of oxygen in arterial blood	462	dopamine	53
systolic blood pressures	441	epinephrine	16
blood platelets	220	dobutamine	3
mean arterial pressure	217	urine output	0

Table 5.4.: Feature Counts in M3

(0.90), certain variables such as *systolic blood pressure* (F1 = 0.50) and *mean arterial pressure* (F1 = 0.67) performed less optimally. These drops may be attributed to greater variability in documentation styles or inconsistencies in value-unit formatting, highlighting areas for further refinement in unit normalization or prompt specificity.

Overall, the hybrid method captures clinically valid measurements with high fidelity and reflects its potential for structured clinical data extraction, especially when supported by symbolic constraints and carefully designed prompt-output pipelines.

Relation to Feature Frequency Extraction performance should be interpreted in relation to feature prevalence (Table 5.4). Frequently documented variables such as *creatinine*, *white blood count*, and *blood platelets* achieved high F1-scores based on substantial sample sizes, supporting the robustness of these estimates. In contrast, variables with near-perfect F1-scores, including *norepinephrine*, *bilirubin*, and *epinephrine*, occur less often. Although performance is strong when these features are present, the smaller number of instances limits statistical certainty.

Lower F1-scores for features such as *systolic blood pressure* and *mean arterial pressure*, which appear more frequently, are therefore more likely attributable to documentation variability or formatting inconsistencies rather than data scarcity. Feature counts thus provide important context for interpreting extraction performance.

1.3 b. What types of clinical variables are not reliably extracted by the hybrid method?

Limitations in Measurement Extraction Despite the strong performance of M3 across many features, several limitations were observed. The model struggled with relative expressions. For example, statements like "platelets above 10" were incorrectly parsed as the exact value 10. This issue was particularly frequent in variables

such as *heart rate*, *mean arterial pressure (MAP)*, and *systolic blood pressure*, where threshold-based descriptions (e.g., "MAP > 70") were misinterpreted as fixed values, resulting in false positives (e.g., 3 false positives for *MAP*).

Approximate or colloquial expressions also contributed to extraction errors. Phrases such as "HR 140s" were interpreted as exact values (i.e., 140), even though they are intended as loose estimates. These issues were especially prevalent in the same cardiovascular variables.

Occasionally, hallucinations occurred despite tight prompt constraints. For example, in the following snippet:

```
Tmax: 36.9 C (98.4 Tcurrent: 36.9 C (98.4 HR: 79 (79 - 83)
```

the model returned a temperature of 99.9, even though this value does not appear anywhere in the text.

Additionally, some domain-specific shorthand expressions were not reliably recognized. For instance, abbreviations like "Tm 100.1" were not identified as temperature values. Ambiguous units (e.g., "2 amps of epinephrine") were also excluded due to uncertainty in dosage or measurement format. When multiple conflicting values were present, the model tended to favor earlier mentions. For example, if EMS documentation listed "Glasgow Coma Score (GCS) 4" but the admission note stated "GCS 3," the earlier value was extracted, highlighting limitations in temporal resolution.

Chunking Omissions and Evaluation Bias While UMLS-driven chunk filtering substantially improved precision by selecting contextually relevant segments, it did not capture all valid mentions. For features like *temperature*, chunks were included if they contained values within a predefined threshold, thereby ensuring relevant mentions were retained. An example of such a chunk is:

```
PHYSICAL EXAMINATION: 98.4, 79/39, 52, 13, 92/3L NC
```

However, for other features, no such numeric heuristics were applied. In these cases, variations in clinical language may have prevented inclusion of valid measurements. For instance, "Urine: 300 mL" was excluded because the term "urine" without the qualifier "output" was neither recognized by the constructed regular expressions nor covered by the UMLS ontology. Consequently, no valid instances of this feature were retained for extraction, resulting in zero successful extractions (Table 5.4).

Since the evaluation was conducted only on preselected chunks, this introduces a form of evaluation bias. Specifically, it inflates the F1 score because only instances with a high prior likelihood of containing a symptom are considered, while alternate phrasings are systematically excluded from evaluation.

Temporal Mistakes and Conflicting Contexts A recurring failure mode involved confusion across temporally distant or conflicting statements within a note. Temporal ambiguity posed a major challenge. Phrases like "Pt received from ED on dopamine

at 10mcgs/kg/min" were often misinterpreted when paired with modifiers such as "meds on hold," leading to incorrect value assignments. Also conflicting contexts occured in different chunks. For example, when an earlier sentence stated "GCS score 4" which was determined when EMS services arrived in the first chunk and a later chunks mentioned "GCS 3, unresponsive," the model often returned the earlier value as it was assigned to stop as soon as a value had been extracted. This suggests that standard transformer-based attention currently may be insufficient for resolving fine-grained temporal conflicts.

1.3 c. Which types of clinical variables are best suited for hybrid extraction?

Feature-Type Suitability The hybrid method was most reliable for concrete, measurable features with standardized formats. Vital signs and laboratory values such as *creatinine*, *bilirubin*, *white blood count*, and *temperature* were extracted with high F1-scores and minimal ambiguity. These features benefit from the explicit value-targeting design of M3, which aligns well with numerical extraction tasks.

In contrast, variables that are more context-dependent or qualitative in nature such as *altered mental status* or *urine output* proved more challenging. In particular, *urine output* was inconsistently extracted when not documented in a standard 24-hour volume format. Similarly, symptom-like indicators that rely on implicit cues or temporal context (e.g., changes in *Glasgow Coma Score*) were more prone to misclassifications or omission. This highlights the difficulty of temporal reasoning, especially in determining what is "current," yet this determination is essential for accurate time-sensitive predictions.

Additionally, variables that are often described using estimates such as *heart rate*, *systolic blood pressure*, and *mean arterial pressure* were frequently extracted as exact measurements, even when phrased approximately (e.g., "HR 140s"). Therefore making them less suitable for feature extraction by M3.

These results suggest that M3 is best suited for quantifiable, well-bounded clinical measurements where surface-level numeric values directly reflect the variable of interest. More nuanced, context-sensitive features may require enhanced temporal modeling or structured clinical context to be reliably extracted.

5.4. Trade-offs Between Coverage, Granularity, and Interpretability

The following research question guides the comparative evaluation of the three extraction methods:

1.4 How do the three approaches (M1, M2 and M3) compare in terms of coverage, granularity, and interpretability?

This overarching question is addressed through the following subquestions:

a. How do the three approaches manage the trade-off between coverage and reliability in clinical feature extraction?

b. In what ways does the hybrid method balance symbolic precision with contextual sensitivity?

c. What trade-offs exist between interpretability, clinician trust, and predictive performance across methods?

These questions guide the comparative analysis of M1, M2, and M3. The evaluation examines trade-offs between coverage, granularity, and interpretability, focusing on how each approach balances reliability with expressive power in clinical feature extraction. Particular attention is paid to differences in symbolic precision, contextual sensitivity, and implications for interpretability and clinician trust. These dimensions are analysed below in relation to subquestions (a)–(c).

1.4 a. How do the three approaches manage the trade-off between coverage and reliability in clinical feature extraction?

M1 (Rule-Based) M1 maximizes precision through deterministic regular expressions. This design minimizes false positives and ensures reliable detections when symptoms are expressed in canonical language. However, recall is extremely limited as lexical variation, abbreviations, and implicit cues are systematically missed. M1 performs poorly in linguistically or temporally complex contexts, as it cannot reason about extract numeric thresholds or temporal ordering such as "patient was febrile yesterday". Statements such as "levophed started" are often misclassified if the exact phrasing is not part of its pattern library. Consequently, M1 strongly favors precision and interpretability at the expense of flexibility, coverage, and contextual understanding.

M2 (LLM Classification) M2 emphasizes coverage and linguistic adaptability. It generalizes across varied clinical language and can infer symptom presence from indirect cues (e.g., interpreting "not responsive" as a low *Glasgow Coma Score*), resulting in very high recall (0.95, Table 5.5). However, this flexibility reduces precision (0.61) and increases false positives, particularly in ambiguous or temporally outdated contexts. Because M2 outputs only continuous confidence scores between 0 and 1 that represent the likelihood that a symptom is present, typically at a thresholded value, it lacks the granularity required to resolve explicit numeric values. Consequently, M2 is well suited for vague, context-dependent variables, but less reliable for precise, measurement-driven features.

M3 (Hybrid Rule-Guided) M3 explicitly aims to balance coverage and reliability by combining symbolic constraints with LLM-based contextual reasoning. Rule-based preselection restricts model input to clinically relevant text segments, reducing

	Data Type	Representation	Acc.	Rec.	Prec.	F1
M1	Binary	Rule-based Flag	-	-	-	-
M2	Continuous	Confidence Score	0.80	0.95	0.61	0.74
M3	Continuous	Raw Measurement Value	0.89	0.94	0.86	0.90

Table 5.5.: Overview of Evaluation Metrics by Method

hallucinations and spurious predictions, while the LLM contributes flexibility in interpreting diverse phrasing and extracting numeric values. This hybrid setup yields both high recall (0.94) and markedly improved precision (0.86), with an overall F1 of 0.90 (Table 5.5). However, this high precision comes at the cost of a potential low sensitivity as the recall is skewed by the filtering stage as relevant information excluded during preselection cannot be recovered and is therefore absent from the evaluation sample. In essence, M3 achieves high precision by minimizing hallucinated outputs, though at the expense of sensitivity, as some true symptoms remain undetected. In practice, M3 performs best for well-defined, quantitative variables (e.g., temperature, creatinine), while remaining less robust for approximate, shorthand, or temporally ambiguous expressions.

Cross-Method Comparison The three methods can be understood as occupying distinct positions along a spectrum of design choices. M1 emphasizes determinism and interpretability, yielding reliable but narrow extraction. M2 prioritizes adaptability and linguistic coverage, but at the cost of reduced precision and temporal accuracy. In addition, M2 produces an output for nearly every processed text segment, whereas M3 generates values only after strict symbolic filtering, resulting in a substantially sparser dataset. M3 occupies an intermediate position: symbolic constraints anchor the extraction process in deterministic rules, while the LLM contributes contextual reasoning and linguistic flexibility.

1.4 b. In what ways does the hybrid method balance symbolic precision with contextual sensitivity?

As observed in RQ1.4a, M3 consistently occupies a middle ground between the narrow precision of M1 and the broad adaptability of M2. This balance arises directly from its hybrid design. By filtering inputs through UMLS-based chunk selection and applying symptom-specific heuristics, the model reduces noise and narrows the scope of text presented to the LLM. In turn, the LLM provides the flexibility to interpret diverse clinical formulations that rigid rules alone would miss.

M3 addresses one of M2's core weaknesses: the tendency to hallucinate values in ambiguous contexts. Constrained input selection lowers hallucination rates and improves measurement fidelity, enabling reliable extraction of structured variables

such as *creatinine*, *bilirubin*, and *white blood count* (often reaching F1-scores $\geq$ 0.90; Table 5.3). Additional safeguards, such as prompt constraints and strict post-processing, further enforce discipline by ensuring that numeric values are only extracted when explicitly supported in the text. At the same time, M3 preserves the main advantage of LLMs: adaptive reasoning. This allows value extraction from non-standard syntax and narrative descriptions, such as correctly interpreting phrases like "creatinine trending down at 1.8" or "white count up to 15."

Despite these advantages, the hybrid design does not fully resolve known challenges. Approximate expressions (e.g., "HR in the 140s") are often misread as exact values, temporal ambiguity persists, and relevant information may be lost during pre-filtering. Moreover, unlike M2 qualitative descriptions without explicit numerics (e.g., "fever," "elevated levels") indicate clinically meaningful thresholds but cannot be captured by M3.

In summary, M3's comparative strength lies not simply in occupying an intermediate position between M1 and M2, but in deliberately guiding flexible language models with clinically grounded constraints. This integration explains its strong performance on structured variables while also clarifying the unresolved limitations in temporal reasoning, approximate expressions, and qualitative descriptors.

1.4 c. What trade-offs exist between interpretability, clinician trust, and predictive performance across methods?

The three approaches differ not only in predictive performance (Table 5.5), but also in how interpretable their outputs are and how strongly clinicians may trust them. Firstly, M1 is the most interpretable: each extraction can be traced to a specific regular expression, making it transparent and auditable. This clarity promotes clinician trust, particularly in high-stakes settings. The trade-off is that performance breadth is limited since coverage is low and nuanced or indirect cues are systematically missed. Then M2 lies at the opposite end of the spectrum. Its contextual flexibility delivers the highest recall and broadest coverage. However, this comes at the expense of interpretability and trust. The decision process is opaque, confidence scores provide little explanation, and false positives in ambiguous or outdated contexts undermine reliability. M3 again occupies a middle ground. Structured prompts, bounded inputs, and post-processing constraints improve transparency compared to M2, making outputs easier to validate. At the same time, performance is much stronger than M1, especially for numeric features, although interpretability is less than that of a purely rule-based system. Clinician trust is therefore higher than for M2 but not as strong as for M1, since hallucination risk is reduced yet not eliminated.

Taken together, the three methods illustrate the inherent tension between predictive performance, interpretability, and clinician trust. M1 and M2 represent opposite extremes: M1 achieves high transparency and strong trust but offers limited predictive breadth, while M2 delivers broad coverage but produces outputs that are difficult to audit and therefore less trustworthy. M3 sits between these poles, combining elements of both approaches. Compared to M1, it substantially improves performance while

giving up some transparency. Compared to M2, it offers more interpretable and trustworthy outputs, though still at the cost of full flexibility. These contrasts show that no method is universally optimal: the appropriate choice depends on whether clinical deployment values reliability and auditability more than maximal detection breadth.

5.5. Limitations

The analyses presented in Sections 5.1–5.4 highlight the relative strengths and weaknesses of the three methods. These findings should, however, be interpreted in light of several limitations that were beyond the scope of this thesis but nonetheless shape the evaluation outcomes:

- **Data Scope:** This study was restricted to physician and nurse notes from MIMIC-III. In addition to the dataset limitations outlined in Section 3.2, the methods were not tested on other note types or in other hospital settings. While these notes are clinically rich and commonly used in retrospective modeling [7], [62] the findings may not generalize to different documentation styles or institutional contexts.

- **Context Window:** The chunk sizes used here (256–512 tokens) were smaller than in related work, which employed windows of 512 [39] and 1000 tokens [37]. Larger windows might improve evaluation metrics, though at the cost of higher computational demand. Moreover, this thesis used sentence boundaries to define chunks, whereas other studies truncated text solely by size. The precise impact of these design choices on performance was not systematically examined.

- **Gold Standard:** In the absence of a fully annotated gold-standard dataset, evaluation relied on proxy metrics, automated scoring, and limited clinician validation. Only a subset of texts was reviewed: 10 per feature, so 190 in total per method of which 42 ambiguous cases in M2 were clinically validated. For M3, all extracted values were annotated by the author. Since M3 only outputs exact measurements, there was less room for interpretive ambiguity. Nevertheless, broader clinical validation would be required to strengthen the reliability and generalizability of these findings.

- **Regular Expressions:** The rule-based expressions were tested only on a small subset of texts, which was insufficient to derive robust evaluation metrics. As a result, they were not refined to their full potential, since exhaustive tuning would have been beyond the scope of this work. This limitation likely influenced the reported performance of the rule-based approach.

6. Methodology: Explainable Sepsis Prediction

6.1. Data Preprocessing

Loading MIMIC-III into PostgreSQL This thesis utilized the MIMIC Code repository provided by the MIT Laboratory for Computational Physiology to load the MIMIC-III dataset into a PostgreSQL database.[1] The 'buildmimic/postgres' scripts were used to create the schema and populate the database with de-identified clinical data, including admissions, procedures, laboratory results, and clinical notes. PostgreSQL was chosen due to its compatibility with the provided scripts and its robust querying capabilities, which are essential for complex cohort selection and data preprocessing.

Loading Data for Model Input To prepare the data for downstream modeling, we adapted the data extraction pipeline used in the MGP-TCN framework[2]. Specifically, based on the 'make query'-based system that automates the SQL querying process, new SQL files have been created to extend the original pipeline. Following the same format as for lab and vital features, notes were extracted 48 hours before the onset of sepsis. These additions ensured that both the case and control cohorts included an additional table linking each subject to their corresponding clinical notes. Unlike the original scripts, chart times were binned in order to keep the same structure across lab, vital and notes to ensure equal chart times after binning.

Sepsis Label Definition The identification of sepsis in this work follows the approach described by Moor et al. (2019) [61], which is based on the Sepsis-3 criteria proposed by Singer et al. (2016) [1]. Organ dysfunction is defined as an increase in the SOFA score by at least two points [52] within a window from 48 hours before to 24 hours after suspected infection. Following the implementation described by Moor et al. (2019) [61], this approach refines and extends the queries developed by Johnson et al. (2016) [70] to generate Sepsis-3 labels on an hourly basis, allowing for the identification of sepsis developing during the ICU stay.

Clinical Note Selection The selection of relevant clinical notes is crucial for accurately capturing patient conditions and clinical decision-making. Whereas previous

[1] `https://github.com/MIT-LCP/mimic-code/tree/main/mimic-iii/buildmimic/postgres`
[2] `https://github.com/BorgwardtLab/mgp-tcn/tree/master`

studies often excluded Discharge Summaries due to their retrospective nature but otherwise included all note types [7], this work restricts the corpus to Physician and Nursing Notes. This choice reduces data volume and computational demands, while focusing on note categories that provide real-time documentation and contain key information about a patient's physiological state and clinical trajectory [88].

Study Cohort This thesis adopted a set of patient inclusion criteria aligned with the approach taken by Moor et al. (2019) [61] to define the study cohort. The cohort comprises patients who were diagnosed with sepsis during their ICU stay (cases) and a corresponding control group. To preserve a realistic class imbalance reflective of clinical settings (approximately 10% sepsis prevalence) the exclusion criteria are followed by the case-control matching procedure (described in the subsequent section 6.1). The inclusion and exclusion criteria are as follows:

- Patients must be over 18 years of age at the time of ICU admission.
- Controls were required to have no history of sepsis, as determined by the absence of ICD codes for sepsis in any prior hospital stay.
- To focus on early sepsis detection, we excluded cases where sepsis onset occurred within the first six hours of ICU admission, due to insufficient early physiological data.
- Patients must have recorded notes (Nursing or Physician Notes) 48h before and 7h after onset of sepsis or target.
- Patient must have developed sepsis 7h after admission such that there is enough data.

After applying these criteria and performing data cleaning and filtering the final study population consisted of 217 sepsis cases and 2304 controls. These slightly alter from the data processing done by Moor et al. (2019) [61] because two new excluding criteria were introduced. Firstly the age was increased to 18 from 14. Secondly, cases were excluded that did not have physician or nursing notes 48 before until 7 hours after the onset of sepsis.

Case-Control Matching To address class imbalance, each sepsis case is matched with ten randomly selected, previously unassigned control patients to get a balanced dataset with 10% sepsis cases and 90% control cases. The control onset time is defined as the same absolute time (measured in hours from ICU admission) at which the matched case met the sepsis criteria. Consequently, for every matched case–control pair, we extract up to 48 hours of data prior to their respective onset times, ensuring all data is drawn after ICU admission.

Variable Selection As explained in Section 4.1 the variables for the unstructured data are based on various sepsis scoring features. (Table A.4 on page iii The variables selected for structured data are the same as selected by Moor et al. (2019) [61] (Table A.9 on page ix). Variables without any measurements within the time window are dropped.

6.2. Pipeline Implementation

Clinical Notes A SQL query was added to the Moor et al. (2019) [61] pipeline [3] to create a materialized view that extracts clinical notes from the MIMIC-III database for ICU stays involving sepsis. Specifically, it retrieves all nursing and physician notes from 48 hours before the recorded sepsis onset time. The result preserves unstructured text data in long format, supporting downstream natural language processing tasks or sepsis timeline analyses. With 217 sepsis cases and 2304 controls resulting in 1452 clinical notes for cases and 13231 clinical notes for controls.

Standardization Continuous variables are standardized using z-scores computed from the training set to ensure consistent feature scaling across train, validation, and test splits. [4] Binary variables from M1, and metadata columns are excluded from standardization. Variables with all missing values in the training set are dropped from all splits.

The standardization follows the z-score formula:

$$z = \frac{x - \mu}{\sigma}$$

where x is the original value, μ is the mean, and σ is the standard deviation calculated from the training set. This ensures comparability across samples and prevents information leakage.

Imputation First all measurements with the same hour and patient id are averaged out leading to a unique measurement per hour and patient. In order to process lack of measurements, the data gets imputated. [5] Following prior work by Li et al. (2024) [7], population-level mean imputation is implemented after forward filling. If a feature did not have any recorded data it was filled with zero.

[3] implemented in `src/query/extract-55h-of-hourly-case-note-series_ex1c.sql` and `src/query/extract-55h-of-hourly-control-note-series_ex1c.sql`

[4] implemented in `src/model/preprocessing/main_preprocessing.py`

[5] implemented in `src/model/preprocessing/bin_and_impute.py`

6.3. Sepsis Prediction Model

Long Short-Term Memory (LSTM) networks, a class of recurrent neural networks (RNNs), are designed to capture temporal dependencies in sequential data [89]. Unlike traditional RNNs, LSTMs incorporate gating mechanisms that allow the model to selectively retain or discard information across long sequences, thereby mitigating the vanishing gradient problem often encountered in deep temporal models. This property makes LSTMs particularly well-suited for multivariate clinical time series, where informative patterns may emerge over variable and irregular time scales.

In this study, a Bidirectional LSTM (BiLSTM) architecture [90] was employed to model patient-specific temporal trajectories within a fixed retrospective window. This design choice was motivated by prior work demonstrating the effectiveness of BiLSTMs for sepsis prediction when incorporating unstructured clinical notes [7]. By processing sequences in both forward and backward temporal directions, the BiLSTM captures not only short-term fluctuations but also longer-range temporal context within a patient's clinical course, enabling more expressive representations of evolving physiological states.

The model was trained to predict sepsis onset at three clinically relevant prediction horizons: 0, 6, and 12 hours before onset. Input data consisted of multivariate sequences of physiological measurements, augmented with LLM-derived feature representations from three proposed methods: M1 (Section 5.1), M2 (Section 5.2), and M3 (Section 5.3). For each experimental configuration, the dataset was split into 80% training, 10% validation, and 10% test sets, with identical partitions used across all model variants to ensure comparability.

Training Procedure The network architecture consisted of a bidirectional LSTM layer with 128 hidden units in each direction, followed by a fully connected output layer producing class logits.[6] Variable-length sequences were padded and masked during batching to ensure that padded time steps did not contribute to the loss. The model was trained using the Adam optimizer with a learning rate of 10^{-3} and a maximum of 100 epochs. Hyperparameters were deliberately chosen to keep training times modest, as the objective of this study was not to maximize predictive performance but to enable a controlled comparison of different information extraction strategies.

Model parameters were optimized by minimizing the cross-entropy loss, yielding probabilistic predictions of sepsis onset. Given the pronounced class imbalance inherent to early sepsis prediction, model selection and early stopping were based on the area under the precision–recall curve (AUPRC) computed on the validation set, rather than validation loss. Early stopping was applied with a patience of 10 epochs to mitigate overfitting.

During evaluation, model outputs were interpreted as calibrated probabilities. Performance was assessed using threshold-independent metrics, with AUPRC serving

[6] Implemented in `src/model/model.py`.

as the primary metric and ROC-AUC reported for completeness. Classification accuracy was computed using a fixed threshold of 0.5 and is included for reference only.

6.4. Explanations

To support post hoc interpretability of the proposed LSTM-based sepsis prediction model, SHAP (SHapley Additive exPlanations) was employed. SHAP values were computed for a representative subset of test sequences at the 6-hour prediction horizon. Specifically, explanations were generated for 10 test sequences, using a background distribution composed of approximately 50 validation sequences. For each explained instance, Shapley values were estimated via Monte Carlo sampling with 100 samples. Specifically, the mean absolute SHAP value was computed for each feature, reflecting its average contribution to the predicted sepsis risk across patients and time.

6.5. Evaluation

Model performance was evaluated using threshold-independent metrics appropriate for highly imbalanced clinical prediction tasks. The primary evaluation metric was the area under the precision–recall curve (AUPRC), which more directly reflects performance on the minority (sepsis) class by jointly capturing precision and recall [61]. In addition, the area under the receiver operating characteristic curve (ROC-AUC) was reported to facilitate comparison with prior work [7], [61]. ROC-AUC measures the model's ability to discriminate between positive and negative cases by quantifying the probability that a randomly selected positive instance is assigned a higher risk score than a randomly selected negative instance. While ROC-AUC is commonly used in sepsis prediction, it may provide an overly optimistic assessment under substantial class imbalance [61]; therefore, it was treated as a secondary metric.

7. Evaluation: Explainable Sepsis Prediction

This section evaluates the impact of integrating text-derived features which are extracted using rule-based (M1), LLM-based (M2), and hybrid (M3) methods into structured sepsis prediction models at 0h, 6h, and 12h prior to onset.

> **RQ 2:** How do features extracted by rule-based, LLM, and hybrid methods differentially affect the performance and interpretability of structured sepsis prediction models?

This overarching research question is addressed through the following subquestions, which separately examine model performance and interpretability:

2.1 How do different text representations (M1-M3) affect predictive performance at different horizons? (Section 7.1)

2.2 Why do the text representations (M1-M3) improve or degrade predictive power? (Section 7.2)

2.3 How does the interpretability of structured sepsis prediction models vary depending on the type of extracted features used (rule-based, LLM, or hybrid)? (Section 7.3)

2.4 What are the effects of improving interpretability on performance? (Section 7.4)

7.1. Performance Results

This section discusses the following research question:

2.1 How do different text representations affect predictive performance at different horizons?

All models share the same architecture and hyperparameters across horizons and input configurations, and were not optimized separately for specific prediction horizons or feature sources. Model performance was evaluated primarily using the area under the precision–recall curve (AUPRC), which is more informative than ROC-AUC under severe class imbalance and better reflects early-warning performance. In this

Horizon (h)	Source	AUPRC	ROC-AUC
0	No Notes	0.652	0.936
	M1	0.612	0.924
	M2	0.608	0.932
	M3	**0.665**	**0.945**
6	No Notes	0.161	0.652
	M1	0.157	0.653
	M2	**0.188**	0.644
	M3	0.137	0.644
12	No Notes	0.273	0.538
	M1	0.176	0.572
	M2	**0.345**	**0.642**
	M3	0.174	0.551

Table 7.1.: Model Performance by Feature Source and Prediction Horizon[1]
[1] Positive class prevalence: 10%.

setting, AUPRC reflects the model's ability to prioritize truly septic patients among a small set of high-risk candidates, which is critical for early warning under alert fatigue and severe class imbalance. ROC-AUC is reported as secondary metrics for comparability with prior work [7], [61]. All results are reported at fixed prediction horizons of 0h, 6h, and 12h prior to sepsis onset.

Results At the 0 hour horizon, all models achieved relatively high AUPRC values, indicating strong predictive performance at onset. As shown in Table 7.1, the hybrid approach (M3) attained the highest AUPRC (0.665), marginally outperforming the baseline model without notes (0.652) as well as the note-based methods M1 (0.612) and M2 (0.608). ROC-AUC values at this horizon were uniformly high (all exceeding 0.92), with M3 again achieving the highest ROC-AUC (0.945). The small performance differences may reflect that structured physiological variables already capture most of the discriminative signal at onset, limiting the incremental contribution of clinical notes.

At the 6 hour prediction horizon, performance declined across all feature configurations, reflecting the increased difficulty of early sepsis prediction under pronounced

class imbalance. AUPRC values were uniformly low, yet meaningful differences emerged: the LLM-based approach (M2) achieved the highest AUPRC (0.188), outperforming both the structured baseline (0.161) and the rule-based method M1 (0.157), while the hybrid model M3 performed worst (0.137). In contrast, ROC-AUC values exhibited minimal variation across models (approximately 0.64–0.65), with marginally higher scores for the baseline and M1. This discrepancy indicates that while ROC-AUC suggests comparable global discrimination, AUPRC reveals substantive differences in minority-class detection, underscoring its greater sensitivity for evaluating early prediction performance in imbalanced settings.

At the 12 hour horizon, performance increased again relative to the 6h setting. The LLM-based method M2 achieved the highest AUPRC (0.345), substantially outperforming the baseline without notes (0.273) and the other note-based approaches (M1: 0.176, M3: 0.174). This trend was mirrored in ROC-AUC, where M2 also achieved the highest value (0.642), compared to 0.538 for the baseline. These results indicate that text-derived features contributed meaningful additional information for longer-term prediction, where structured variables alone were less informative.

Horizon-Dependent Effects of Text-Derived Features In summary, AUPRC-based evaluation demonstrates that the benefit of incorporating clinical notes is strongly horizon-dependent. While performance gains at onset were modest, the LLM-based approach (M2) consistently yielded the largest improvements at earlier prediction horizons, particularly 12 hours before onset. Cohorts differed across horizons due to minimum length-of-stay requirements, and a uniform imputation strategy was applied across all models to ensure comparability. These factors may partially explain the monotonic degradation in ROC-AUC observed with increasing prediction horizon, although such a pattern was not reported by Moor et al. (2019) [61], despite their use of a broadly similar data pipeline.

Importantly, incorporating clinical notes did not uniformly improve predictive performance: rule-based features and constrained hybrid extraction degraded performance at earlier horizons, indicating that text-derived information can be uninformative or even harmful when misaligned with the prediction context. Crucially, these results demonstrate that the inclusion of unstructured text features does not guarantee performance gains and may actively degrade model performance when extracted signals are noisy, redundant, or poorly matched to the prediction horizon.

7.2. Interpretation of Method Differences

This section answers the following research question:

2.2 Why do the text representations (rule-based, LLM, and hybrid) improve or degrade predictive power?

This section interprets the performance differences observed in Section 7.1 by

examining how each extraction strategy represents information from unstructured clinical notes.

Limits of Presence-Based Encoding (M1) The rule-based approach (M1) encodes information from clinical notes as binary indicators of whether specific measurements are mentioned, discarding severity, directionality, and temporal context (see Section 5.1). Although the underlying concepts captured by M1 occur more frequently in septic patients, collapsing these signals into presence-only features results in substantial information loss and redundancy with structured variables. In clinical narratives, predictive value typically lies not in the mention of a measurement but in how it is described and contextualized. For example, references to altered mental status via the Glasgow Coma Scale (GCS) may indicate concern, yet without encoding degree or trajectory of impairment, such mentions remain too coarse to support early discrimination. The weak performance of M1 across horizons is therefore consistent with representational limitations rather than an absence of signal in the notes.

Contextual Inference (M2) In contrast, the LLM-based approach (M2) leverages narrative context to infer latent clinical states that are not explicitly quantified. Rather than extracting exact measurements, M2 maps descriptive language onto structured features that approximate underlying physiological or cognitive states. Using GCS as an illustrative example, M2 may infer impaired mental status from qualitative descriptions even when no numeric score is documented. This form of contextual approximation is plausibly most informative at earlier prediction horizons (6–12 hours before onset), when structured measurements are sparse but narrative documentation already reflects emerging concern. Importantly, this interpretation does not assume that inferred states are always accurate, only that they are sufficiently correlated with future sepsis onset to improve early ranking performance. This behavior directly reflects the intrinsic properties observed in Section 5.2, where M2 achieved very high recall (0.95) by inferring symptom presence from indirect or qualitative descriptions. While this led to reduced precision at the extraction level, these weak but contextually grounded signals appear to be particularly valuable at earlier horizons, when structured physiological measurements are sparse and narrative documentation already reflects emerging clinical concern.

Precision Bias, Redundancy, and Noise in Hybrid Extraction (M3) The hybrid extraction strategy (M3) emphasizes precision by constraining outputs to explicitly documented, numerically grounded variables through strict prompting and post-processing. While this improves traceability and is advantageous when sepsis is clinically manifest, it systematically filters out weak signals that may be informative earlier in the disease trajectory. In addition, many M3-derived variables overlap with structured inputs, limiting their complementary value. In early prediction settings where signal is weak, such redundancy can dilute or suppress complementary information under regularization, an effect that may be exacerbated by noisy or

temporally ambiguous documentation. These factors provide a plausible explanation for why M3 improves performance at onset but degrades performance at earlier horizons, although alternative explanations such as feature sparsity (see Table 5.4) or cohort effects cannot be excluded. This pattern aligns with the intrinsic evaluation in Section 5.3, which showed that M3 achieves high extraction precision by restricting outputs to explicitly documented numerical values. While this improves fidelity at onset, it systematically filters out weak or implicit signals. At earlier horizons, where documentation could be more narrative and less quantitative, this filtering suppresses potentially informative context, leading to degraded predictive performance.

Horizon-Aligned Representational Trade-offs Taken together, these observations suggest that the utility of text-derived features depends on the alignment between representational assumptions and prediction horizon. Early sepsis prediction benefits from expressive, context-sensitive representations capable of capturing weak narrative signals (M2), whereas late-stage prediction favors precise, explicitly grounded extraction (M3). When this alignment is mismatched, incorporating clinical notes can be neutral or even detrimental to performance. This horizon-dependent trade-off provides a unifying explanation for the divergent behavior of M1, M2, and M3 and motivates further analysis of feature-level contributions and comparison with prior work.

Results in Context of Related Work Prior studies report higher AUPRC and ROC-AUC than those achieved in this work [7], [61]. However, these approaches primarily focus on end-to-end model optimisation, whereas the present study investigates alternative strategies for extracting and representing information from unstructured clinical text. In contrast to prior work, interpretable feature representations are evaluated. This design enables a systematic analysis of when text-derived features contribute predictive value, when they are redundant or detrimental, and how interpretability constraints influence performance. Consequently, the primary contribution of this study lies not in numerical performance gains, but in providing mechanistic insight into the representational trade-offs that shape the effectiveness of LLM-based text extraction for early sepsis prediction.

7.3. Interpretability Analysis (SHAP)

2.3 How does the interpretability of structured sepsis prediction models vary depending on the type of extracted features (rule-based, LLM, or hybrid) used?

To address this question, interpretability is examined through the lens of post-hoc feature attribution. Using SHAP values at a representative 6-hour prediction horizon, the analysis compares how rule-based, LLM-based, and hybrid features alter the model's reliance on structured and text-derived variables. Beyond characterizing

these differences, the analysis also highlights the implications of feature design choices for traceability and clinical verifiability.

Feature Influence Across Methods The SHAP analyses at the 6-hour prediction horizon reveal systematic differences in how each extraction strategy influences model behavior, highlighting the importance of aligning text-derived feature representations with horizon-specific clinical documentation. Importantly, interpretability in this context refers to the traceability and clinical verifiability of influential features, rather than to the correctness of the underlying predictions. In the baseline model without clinical notes, predictive importance is dominated by structured physiological and laboratory variables such as *fibrinogen, heart rate, temperature, respiratory rate, and white blood cell count* (Figure B.4a on page xiv). This pattern indicates that, in the absence of textual input, the model relies primarily on acute physiological markers that are already well captured in structured form.

Although the rule-based approach (M1) introduces additional text-derived features, the SHAP rankings (Figure B.4b on page xiv) show that these features are structured variables (e.g., *oxygen flow, temperature, heart rate, mean arterial pressure*). Consequently, M1-derived features do not meaningfully reshape the model's importance structure but instead redistribute importance across highly correlated variables. This pattern suggests that rigid presence-based encodings primarily amplify redundancy and correlation without adding discriminative information. While the ordering of top features differs from the baseline model, these shifts largely reflect reweighting effects rather than genuine representational gains, helping to explain why M1 fails to improve predictive performance.

In contrast, the LLM-based approach (M2) introduces several text-derived variables from the extraction pipeline that rank among the most influential predictors (Figure B.5a on page xv). Features such as *norepinephrine, epinephrine, GCS, tachypnea, and bilirubin* appear alongside core physiological variables from structured data, including *heart rate* and FiO_2. This expanded and heterogeneous feature set reflects associations with higher-level clinical context and treatment patterns embedded in narrative documentation, which is consistent with the superior predictive performance of M2 at earlier horizons. However, this increased expressiveness comes at a cost: many influential M2-derived features are inferred rather than explicitly measured, making their contributions harder to verify against source documentation and more sensitive to contextual ambiguity. As discussed in Section 5.2, features such as *GCS* exemplify this trade-off, as they gain predictive influence through contextual inference while simultaneously reducing auditability and interpretive certainty.

The hybrid approach (M3) occupies an intermediate position between the rule-based and LLM-based methods. Its SHAP profile (Figure B.5b on page xv) contains fewer text-derived features overall, with only one clinically grounded variable (*creatine*) entering the top-ranked predictors alongside standard physiological measurements. Notably, creatinine is also among the most frequently extracted features (Table 5.4), suggesting that its prominence in the model is supported by both reliable extrac-

tion and substantial representation in the dataset. While this anchoring improves traceability and clinical plausibility relative to M2, the restricted feature set limits the model's ability to exploit broader contextual signals present in narrative notes. This observation is consistent with the intrinsic evaluation in Section 5.3, which highlighted the sparsity and strict filtering imposed by the hybrid extraction pipeline. At the 6-hour horizon, this constraint appears to suppress potentially informative signals, helping to explain the degradation in performance observed for M3 relative to M2.

Interpretability and Traceability While SHAP values indicate which features most strongly influence model predictions, they do not imply clinical causality or physiological ground truth. In this setting, feature importance reflects predictive relevance within the model rather than the correctness or reliability of the extracted clinical information. Text-derived features may therefore capture patterns of clinical documentation and clinician concern that are predictive of downstream outcomes, even when they do not correspond to explicit measurements. Despite substantial differences in intrinsic interpretability across methods (Section 5.4), all approaches remain traceable, as the text segments underlying each extracted feature can be inspected and audited.

Taken together, the SHAP analyses indicate that at the 6-hour prediction horizon, the most interpretable method, M3, contributes relatively few influential features compared to the LLM-based approach (M2). In the case where structured physiological signals are limited, M2 benefits from more expressive text-derived representations that introduce additional predictive signals, albeit with reduced verifiability. This observation motivates a closer examination of horizon-dependent trade-offs between expressiveness and interpretability.

7.4. Balancing Performance and Interpretability

2.4 What are the effects of improving interpretability on performance?

At earlier prediction horizons, increasing interpretability acts as a constraint on the type of information available to the model. As shown in Table 7.1, methods prioritizing traceability and clinical verifiability (M1, M3) are most effective at sepsis onset, whereas more expressive LLM-based representations (M2) become advantageous at earlier horizons, when structured physiological measurements are sparse.

Rule-based features (M1) constitute the most interpretable extraction strategy, as each feature corresponds to a deterministic pattern match that is fully traceable to the source text. This design maximizes transparency and auditability, but severely restricts representational flexibility. By collapsing narrative information into binary presence indicators, M1 privileges explicit mentions over implicit clinical reasoning and introduces substantial redundancy with structured variables. As a result, this

approach limits the model's capacity to leverage weak or emerging signals in narrative documentation, which are particularly relevant at earlier prediction horizons.

At the opposite end of the spectrum, LLM-derived features (M2) allow the model to incorporate broader contextual information from clinical narratives, including treatment patterns and qualitative assessments that may precede overt physiological deterioration (Section 5.2). This expressiveness expands the space of admissible evidence available to the model at early horizons, but comes at the cost of reduced intrinsic interpretability and verifiability. Many influential M2-derived features are inferred rather than explicitly documented, making their provenance harder to validate at the individual prediction level and increasing sensitivity to contextual ambiguity. Thus, M2 trades strict interpretability for access to early, narrative-driven signals.

The hybrid approach (M3) represents a deliberate compromise between these extremes. By constraining extraction to explicitly documented, numerically grounded measurements, M3 improves feature-level traceability and clinical plausibility relative to M2, while retaining limited contextual flexibility. This design aligns well with prediction at onset, when abnormalities are more likely to be explicitly recorded. However, these same constraints reduce the model's ability to exploit weak or implicit narrative cues at earlier horizons, effectively biasing the representation toward precision over signal discovery, as discussed in Section 5.3.

Taken together, these findings indicate that improving interpretability through stricter extraction constraints does not uniformly enhance model utility, particularly at earlier prediction horizons. Designing interpretable sepsis prediction systems therefore requires conscious, horizon-dependent decisions about whether to privilege explicit, verifiable measurements or to admit more ambiguous but potentially informative narrative signals. Rather than treating interpretability as an absolute objective, it must be balanced against the informational demands of early warning, where uncertainty is inherent and signals are often indirect.

7.5. Limitations

The analyses presented in Sections 7.1–7.4 highlight the strengths of incorporating each of the three methods in a sepsis prediction model. These findings should, however, be interpreted in light of several limitations that were beyond the scope of this thesis but nonetheless shape the evaluation outcomes:

- **Model Assumptions and Generalizability:** FLAN-T5 and similar instruction-tuned LLMs have not yet been extensively evaluated in real-world clinical workflows. Their outputs may reflect pretraining biases, and prompt-based extraction does not guarantee robustness across institutions. Comprehensive ethical and clinical validation will therefore be essential prior to deployment.

- **False Interpretability:** While structured outputs (e.g., JSON schema) and SHAP visualizations improve transparency, they can also create a false sense

of reliability, particularly when features are inferred by LLMs from ambiguous or narrative text rather than directly observed measurements.

- **Temporal Resolution:** As discussed in Section 5, the temporal resolution of M1, M2, and M3 remains limited. Such inconsistencies reduce extraction fidelity and undermine downstream model interpretability, since accurate time-stamped data is essential for sepsis prediction.

- **Sparse Feature Coverage:** Because M3 only extracted explicitly documented measurements, coverage was sparse for several variables. For example, `M3_urine` yielded no extractions. Many extracted variables were either too infrequent or redundant with structured inputs to meaningfully affect predictions.

- **Lower Predictive Performance Compared to Prior Work:** The ROC-AUC and PRAUC obtained in this study are substantially lower than those reported in earlier work using more advanced temporal architectures such as MGP-TCN [61]. This suggests that future evaluations should test the integration of extracted features alongside stronger baseline models to assess whether performance gains hold in more capable frameworks.

- **Dataset Limitations:** The MIMIC-III dataset, while widely used for benchmarking, is dated and reflects a single-center cohort. It was selected here to ensure comparability with prior sepsis prediction models such as MGP-TCN [61]. However, using more recent and diverse datasets would improve the relevance and generalizability of findings to current EHR systems.

8. Conclusion

8.1. Summary

This thesis investigated whether interpretable methods for extracting clinically relevant features from unstructured clinical notes can improve early sepsis prediction. Three extraction strategies were developed and evaluated using the MIMIC-III database: a deterministic rule-based system (M1), an LLM-based extraction approach (M2), and a hybrid rule-guided quantification model (M3). These methods were integrated into a structured sepsis prediction pipeline and assessed with respect to both predictive performance and interpretability across multiple prediction horizons.

The results demonstrate that the utility of text-derived features is strongly horizon-dependent. While structured physiological variables dominate prediction at onset, incorporating clinical notes becomes increasingly beneficial at earlier horizons. In particular, LLM-based extraction (M2) provided the largest performance gains when predicting sepsis further from onset, whereas more constrained and interpretable approaches showed limited or inconsistent benefits outside the immediate onset window. In this sense, interpretability operates as a representational filter, trading expressive capacity for verifiability in ways that become particularly consequential when early detection relies on indirect narrative cues.

Key Findings and Contributions. This thesis addressed the following research questions:

1. **RQ1:** How accurately can rule-based methods, LLMs, and hybrid models extract sepsis-related clinical features from unstructured clinical notes?

2. **RQ2:** How do features extracted by these methods differentially affect the performance and interpretability of structured sepsis prediction models?

Regarding RQ1, the findings indicate that rule-based methods (M1) ensure full auditability but capture only a limited range of linguistic expressions. LLM-based extraction (M2) identifies more implicit and context-dependent clinical cues, but introduces hallucinations and reduced traceability. Nevertheless, M2 performs well for variables that rely on broader contextual interpretation rather than explicit measurements. The hybrid approach (M3) mitigates some of these risks by anchoring extracted features to explicit measurements, resulting in higher precision but reduced flexibility when note content deviates from expected formats. M3 performs

particularly well when features are documented explicitly and with exact numeric values. Overall, this thesis demonstrates that extraction accuracy is fundamentally constrained by a trade-off between linguistic coverage and auditability, with no single method consistently dominating across all clinical contexts.

Regarding RQ2, integrating these features into structured prediction models revealed clear trade-offs. Rule-based features contributed minimal predictive value across horizons. LLM-derived features yielded the strongest improvements at earlier horizons, particularly 12 hours prior to onset, where structured signals were weakest. In contrast, the hybrid approach achieved its best performance at onset but degraded performance when applied further from onset, suggesting sensitivity to horizon-specific documentation styles. These results indicate that increased interpretability does not uniformly improve predictive performance and may constrain useful contextual information when applied indiscriminately.

Main Contributions Taken together, this thesis makes the following contributions:

1. **Feature extraction methodologies:** The development of three complementary approaches for extracting sepsis-related information from unstructured clinical notes, spanning fully transparent rule-based extraction, flexible LLM-based extraction, and a novel hybrid rule-guided quantification strategy.

2. **Horizon-aware evaluation:** A systematic comparison demonstrating that the effectiveness of text-derived features depends critically on prediction horizon, with LLM-based features providing the greatest benefit at earlier time points.

3. **Performance–interpretability analysis:** Empirical evidence that improving interpretability does not guarantee improved performance, highlighting the need for horizon- and task-specific trade-offs.

4. **Reproducible pipeline:** An end-to-end, reproducible framework for integrating unstructured clinical text into structured prediction models, supporting transparent and extensible clinical AI research (see Appendix C).

Outlook Beyond sepsis prediction, this work suggests that interpretability in clinical machine learning should not be treated as a fixed property, but as a task- and horizon-dependent design decision that shapes representational capacity. Models that prioritize strict traceability may enhance verifiability, but at the potential cost of excluding weak or narrative-driven signals that are essential for early detection. Effective clinical AI systems therefore require deliberate calibration of interpretability against informational demands. These findings highlight the need for further methodological refinement and broader validation, which are discussed in Section 8.2.

8.2. Future Work

Future research could extend this work along several focused directions. A key methodological avenue is the tighter integration of LLM-based extraction (M2) with rule-guided quantification (M3). Combining retrieval-augmented generation for contextual grounding with constrained extraction for numerical precision may reduce hallucinations while preserving linguistic flexibility, albeit at increased computational cost. Such an approach would enable feature-level decisions regarding when contextual inference is preferable to deterministic extraction. In parallel, future work should systematically investigate which types of variables benefit most from specific extraction strategies, particularly for features that are highly context-dependent, and evaluate extraction performance on a per-variable basis.

A second direction concerns data coverage and generalizability. This study focused primarily on physician and nurse notes, yet other documentation sources—such as progress notes, emergency department reports, and radiology impressions—contain complementary clinical signals. Prior work has shown that incorporating a broader range of note types improves sepsis prediction performance [7]. Evaluating extraction strategies across note types and institutions is therefore essential for assessing robustness and transferability. Furthermore, systematic analysis is needed to identify clinically relevant features that are consistently documented in free-text notes but absent from structured EHR fields, as these signals represent a key opportunity for additional predictive gain.

Finally, expanded clinical validation remains critical. Larger gold-standard datasets, supported by human-in-the-loop annotation and active learning, would improve evaluation reliability while limiting annotation burden. Real-time evaluation across diverse care settings will ultimately be necessary to establish clinical utility, robustness, and user trust.

Future Research Questions Based on these directions, several focused research questions emerge:

- **FRQ1:** How can retrieval-augmented generation and extraction-based quantification be effectively combined to balance contextual grounding and numerical precision?
- **FRQ2:** Which features are better captured by generative versus extractive approaches, and how can feature-level evaluations guide selective method deployment?
- **FRQ3:** Which clinical note types (e.g., progress notes, emergency department reports, radiology impressions) contribute most to predictive performance, and how does their inclusion affect cross-institutional generalizability?
- **FRQ4:** Which clinically relevant signals appear in unstructured notes but are not represented in structured EHR data and are suitable for automated extraction?

- **FRQ5:** How does real-time deployment across diverse clinical settings affect model robustness, clinical utility, and clinician trust?

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A. Tables

A.1. Sepsis Scoring (Section 4.3)

System	0	1	2	3	4
Respiration PaO_2/FiO_2 (torr)	>400	≤400	≤300	≤200 with respiratory support	≤100 with respiratory support
Coagulation Platelets ($\times 10^3/mm^3$)	>150	≤150	≤100	≤50	≤20
Liver Bilirubin (mg/dL) (μmol/L)	<1.2 (<20)	1.2–1.9 (20–32)	2.0–5.9 (33–102)	6.0–11.9 (103–204)	>12 (>204)
Cardiovascular Hypotension	No hypotension	MAP <70 mmHg	Dopamine ≤5 or dobutamine (any dose)[a]	Dopamine >5 or epi ≤0.1 or norepi ≤0.1[a]	Dopamine >15 or epi >0.1 or norepi >0.1[a]
Central Nervous System Glasgow Coma Score	15	13–14	10–12	6–9	<6
Renal Creatinine (mg/dL) (μmol/L) or urine output	<1.2 (<110)	1.2–1.9 (110–170)	2.0–3.4 (171–299)	3.5–4.9 (300–440) or <500 mL/day	≥5.0 (>440) or <200 mL/day

epi, epinephrine; norepi, norepinephrine.
[a] Adrenergic agents administered for at least 1 hr (doses in μg/kg/min).
To convert torr to kPa, multiply by 0.1333.

Table A.1.: SOFA Score Table from Vincent et al. [52]

Criterion	Threshold
Temperature	$> 38°C$ or $< 36°C$
Heart Rate	> 90 beats per minute
Respiratory Rate	> 20 breaths per minute or $PaCO_2 < 32$ mmHg
White Blood Cell Count	$> 12{,}000/mm^3$, $< 4{,}000/mm^3$, or $> 10\%$ immature (band) forms

Table A.2.: SIRS Criteria [51]

Criterion	Threshold
Respiratory Rate	≥ 22 breaths per minute
Altered Mentation	Glasgow Coma Scale (GCS) < 15
Systolic Blood Pressure	≤ 100 mmHg

Table A.3.: qSOFA Score Criteria [1]

A.2. Variables Unstructured Data (Section 4.3)

Name	UMLS name	UMLS ID	Vocab.	Vocab. ID	Scoring
Dobutamine	dobutamine	C0012963	RxNorm	1337720	SOFA-score
Dopamine	dopamine	C0013030	RxNorm	1337860	SOFA-score
Epinephrine	epinephrine	C0014563	RxNorm	1343916	SOFA-score
Norepinephrine	norepinephrine	C0028351	RxNorm	1321341	SOFA-score
Glasgow Coma Scale	Glasgow coma score.total	C0806223	LOINC	3007194	SOFA, qSOFA
Bilirubin	Bilirubin.total [Mass/volume] in Serum or Plasma	C0364108	LOINC	3024128	SOFA
Creatinine	Creatinine [Mass/volume] in Serum or Plasma	C0364294	LOINC	3016723	SOFA
Oxygen Partial Pressure	partial pressure of oxygen in arterial blood	C1822073	LOINC	3027801	SOFA
Platelets in Blood	Blood Platelets	C0005821	LOINC	3007461	SOFA
Urine output	Rate of urine output, function	C0232856	LOINC	3014315	SOFA
Artificial ventilation	Respiration, Artificial	C0035205	LOINC	4230167	SOFA
Mean arterial pressure	Mean arterial pressure (observable entity)	C5780931	SNOMED	37168599	SOFA
Fever	Fever	C0015967	SNOMED	437663	SIRS
Hypothermia	Hypothermia, natural	C0020672	SNOMED	435371	SIRS
Tachycardia	Tachycardia	C0039231	SNOMED	444070	SIRS
Tachypnea	Tachypnea	C0231835	SNOMED	40478545	SIRS, qSOFA
Leukocytosis	Leukocytosis	C0023518	SNOMED	438398	SIRS
Leukopenia	Leukopenia	C0023530	SNOMED	435224	SIRS
Systolic Blood Pressure	Systolic blood pressure measurement	C1306620	SNOMED	4152194	qSOFA

Table A.4.: UMLS concepts and vocabulary mappings used for unstructured clinical variables.

A.3. M1: Regular Expressions (Section 4.5)

CONCEPT_NAME	RegEx
Dobutamine	`\b(dobutamine|dobutamin)\b`
Dopamine	`\b(dopamine|dopamin|da)\b`
Epinephrine	`\b(epinephrine|epinephrin|adrenaline|adrenalin|epi)\b`
Norepinephrine	`\b(norepinephrine|ne|noradrenaline|noradrenalin|norepi|nor-epi)\b`
Glasgow Coma Scale ≤ 14	`\b(gcs|glasgow[ -]?coma[ -]?scale|glasgow[ -]?coma)\b`
Bilirubin ≥ 1.2 mg/dL	`\b((tot(al)?)[ -]?)?(bilirubin|bili)([ -]?(tot(al)?))?\b[-]?(\d+(\.\d+)?\b)?`
Creatinine ≥ 1.2 mg/dL	`\b((creatinine|cr|cre|creat|kreatinine|kre|k|kreat)[.]?)\b[-]?(\d+(\.\d+)?\b)?`
Oxygen $PaO_2/FiO_2 \leq 400$	`\b(pao2|po2|Pao_2|p[ -]?ao2)\b`
Platelets $\leq 150{,}000/\mu$L	`\b((platelets|thrombocytes|plt|plts|thrombozyten)[-]?(ct|cnt|count))\b[-]?(\d+(\.\d+)?\b)?`
Urine output < 500 mL/day	`\b(urine[-]?output|u[-]?output|urinary[ -]?output|uop)\b`
Artificial ventilation	`\b(ventilator|ventilation|mechanical[-]?ventilation|respirator|PEEP|FiO2|tidal[ -]?volume|TV|respiratory[-]?rate|RR|Minute[-]?Volume|MV|inspiratory[ -]?pressure|IPAP|expiratory[-]?pressure|EPAP|APRV|BiPAP|CPAP)\b`
Mean arterial pressure $<$ 70 mmHg	`\b(map|mean[ -]?arterial[ -]?pressure)\b`
Fever > 100.4°F	`\b(fever|pyrexia|\b(100[4-9]|10[1-9][0-9]|11[1-9][0-9]|10[1-9]|11[0-9])\b[ ]?(F|degrees)|T[ -]?(100[4-9]|10[1-9][0-9]|11[1-9][0-9]|10[1-9]|11[0-9]))\b`
Hypothermia < 96.8°F	`\b(hypothermia|\b(96[0-8]|9[0-5][0-9]|[1-8][0-9]|9[0-6]|[1-8][1-9][0-9])\b[ ]?(F|degrees))\b`
Tachycardia > 90 bpm	`\b(tachycardia|(heart[ ]?rate|HR)[ ]\b([9][0-9]|1[0-9][0-9]|2[0-9][0-9])\b)\b`
Tachypnea > 20 bpm	`\b(tachypnea|(breathing[ -]?rate|respiratory[ -]?rate|respirations)[ ]\b([2-9][0-9])\b)\b`
Leukocytosis $> 12{,}000/\mu$L	`\b(leukocytosis|leukokytosis|\b(1[2-9][0-9][0-9][0-9]|[2-9][0-9][0-9][0-9][0-9])/μL|WBC[ -]?\b([1][4-9][0-9]|[2-9][0-9][0-9])\b)\b`
Leukopenia $< 4{,}000/\mu$L	`\b(leucocytopenia|leucopenia|leukocytopenia|leukopenia|\b([1-9]|[1-9][0-9]|[1-9][0-9][0-9]|[1-3][0-9][0-9][0-9])/μL|WBC[ -]?\b([1-3][0-9])\b)\b`

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CONCEPT_NAME	RegEx
Systolic blood pressure ≤ 100 mmHg	`\b(sap|systolic[ -]?arterial[ -]?pressure)\b`

Table A.5.: Regular expression patterns used to identify clinical concepts in text.

A.4. M2: Prompt Questions per Symptom (Section 4.6)

Feature	M2_question
Dobutamine	Does this patient currently have any dose of dobutamine?
Dopamine	Does this patient currently have any dose of dopamine?
Epinephrine	Does this patient currently have any dose of epinephrine?
Norepinephrine	Does this patient currently have any dose of norepinephrine?
Glasgow coma score	Does the patient currently have a score 14 or lower on the Glasgow Coma Scale?
Bilirubin	Does the patient currently have a bilirubin level greater than 1.2 milligrams per deciliter?
Creatinine	Does the patient currently have a creatinine level greater than 1.2 milligrams per deciliter?
Partial Pressure of Oxygen in Arterial Blood	Does this patient currently have PaO_2/FiO_2 ratio below 400 mm Hg?
Blood Platelets	Does this patient currently have a platelet count lower than 150 thousand per microliter?
Urine Output	Is the patient's current urine output less than 500 milliliters per day?
Artificial Respiration	Is the patient currently receiving artificial ventilation?
Mean arterial pressure	Is the patient's current mean arterial pressure less than 70 mmHg?
Fever	Does the patient currently have a fever greater than 100.4 F?
Hypothermia	Does the patient currently have hypothermia, defined as a temperature less than 96.8 F?
Tachycardia	Does the patient currently have tachycardia, with a heart rate greater than 90 beats per minute?
Tachypnea	Does the patient currently have tachypnea, with a respiratory rate greater than 20 breaths per minute?

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Feature	M2_question
Leukocytosis	Does the patient currently have leukocytosis, with a white blood cell (WBC) count greater than 12.0E9/L?
Leukopenia	Does the patient currently have leukopenia, with a white blood cell (WBC) count less than 4.0E9/L?
Systolic blood pressures	Does the patient currently have a systolic blood pressure of 100 mm Hg or less?

Table A.6.: Clinical concept questions for each feature.

A.5. M3: Prompt Questions and Examples per Symptom (Section 4.6)

Name	M3_question
Dobutamine	What is the current dose of dobutamine being administered (μg/kg/min)?
Dopamine	What is the current dose of dopamine being administered (μg/kg/min)?
Epinephrine	What is the current dose of epinephrine being administered (μg/kg/min)?
Norepinephrine	What is the current dose of norepinephrine being administered (μg/kg/min)?
Glasgow coma score	What is the patient's current Glasgow Coma Scale (GCS) score?
Bilirubin	What is the patient's current bilirubin level (mg/dL)?
Creatinine	What is the patient's current creatinine level (mg/dL)?
Partial Pressure of Oxygen in Arterial Blood	What is the patient's current PaO/FiO ratio?
Blood Platelets	What is the patient's current platelet count ($\times$10/L)?
Urine Output	What is the patient's current urine output (mL) per day?
Artificial Respiration	Is the patient receiving artificial ventilation?
Mean arterial pressure	What is the patient's current mean arterial pressure (mmHg)?
Temperature	What is the patient's current temperature (Fahrenheit)?
Tachycardia	What is the patient's current heart rate (beats per minute)?

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Name	M3_question
Tachypnea	What is the patient's current respiratory rate (breaths per minute)?
Leukocytosis	What is the patient's current white blood cell count ($\times 10^9$/L)?
Leukopenia	What is the patient's current white blood cell count ($\times 10^9$/L)?
Systolic blood pressures	What is the patient's current systolic blood pressure (mmHg)?

Table A.7.: M3: Clinical concept questions for each feature

Name	Example (Text + Output)
Dobutamine	Text: "Dobutamine - 8 mcg/Kg/min" Output: {{"value": 8}} Text: "Patient required norepinephrine, dobutamine, and milrinone." Output: {{"value": null}}
Dopamine	Text: "IV dopamine initiated @ 5mcg/kg/hr and quickly titrated to 15mcg/kg/min." Output: {{"value": 15}} Text: "She arrived in the CCU on dopamine 10mic/kilo" Output: {{"value": 10}} Text: "Dopamine - 5 mcg/Kg/min" Output: {{"value": 5}} Text: "Dopamine and Levophed gtts started" Output: {{"value": null}}
Epinephrine	Text: "Epi started at 0.02mcg/kg/min" Output: {{"value": 0.02}} Text: "wean Epi for CI > 2.0" Output: {{"value": null}}
Norepinephrine	Text: "Norepinephrine - 0.06 mcg/Kg/min" Output: {{"value": 0.06}} Text: "11. Norepinephrine 12. Pneumococcal Vac Polyvalent" Output: {{"value": null}}
Glasgow coma score	Text: "GCS = 6" Output: {{"value": 6}} Text: "current GCS scale of 9 (< 10)" Output: {{"value": 9}}
Bilirubin	Text: "patient's bilirubin was 6" Output: {{"value": 6}} Text: "bili to 7.0" Output: {{"value": 7}} Text: "Bili Neg" Output: {{"value": null}}
Creatinine	Text: "Cr 2.5" Output: {{"value": 2.5}} Text: "Creatinine up to 1.8" Output: {{"value": 1.8}}

Continued on next page

Name	Example (Text + Output)
Partial Pressure of Oxygen in Arterial Blood	Text: "PaO2 / FiO2: 602" Output: {{"value": 602}} Text: "PaO2 / FiO2: 142" Output: {{"value": 142}}
Blood Platelets	Text: "Plt 129" Output: {{"value": 129}} Text: "plt 101" Output: {{"value": 101}} Text: "pt transfused with 1 unit platelets" Output: {{"value": null}}
Urine Output	Text: "uop 450ml" Output: {{"value": 450}} Text: "Lasix gtt 10–15 mg/hr with goal 100–150 cc/hr of UOP" Output: {{"value": null}} Text: "Urine output 4–20cc clear concentrated urine hourly." Output: {{"value": null}}
Mean arterial pressure	Text: "Levophed @ .14mcg/kg/min to maintain MAP 65" Output: {{"value": 65}} Text: "wean Epi for CI > 2.0 and MAP > 60" Output: {{"value": null}} Text: "Vasopressin also initiated @ 2.4 units/hr for MAP goal ¿60" Output: {{"value": null}}
Temperature	Text: "T current: 37.7 (99.9°F)" Output: {{"value": 99.9}} Text: "Constitutional: No fevers" Output: {{"value": null}} Text: "Febrile to 103.1." Output: {{"value": 103.1}}
Heart Rate	Text: "HR: 66 (66 - 88) bpm" Output: {{"value": 66}} Text: "BP nadir 69/42 with HR 98." Output: {{"value": 98}} Text: "HR 70s-90s NSR with no ectopy noted." Output: {{"value": null}}
Respitory Rate	Text: "RR: 21 (15 - 21) insp/min" Output: {{"value": 21}} Text: "She meets SIRS criteria with leukocytosis, tachycardia, and tachypnea" Output: {{"value": null}}
WBC	Text: "WBC 16.4" Output: {{"value": 16.4}} Text: "LP showed 9500 WBC (91% polys)" Output: {{"value": 9500}} Text: "Patient with elevated WBC count" Output: {{"value": null}}
Systolic Blood Pressure	Text: "SBP close to 100 tonight..." Output: {{"value": 100}} Text: "SBP stable around 102" Output: {{"value": 102}} Text: "SBP in the ED at [**Hospital 2437**] Hospital was in the 80's." Output: {{"value": null}}

Table A.8.: M3: Example value extraction instances per feature

A.6. Variables Structured Data (Section 6.1)

Category	Parameters
Vital Parameters	Systolic Blood Pressure, Diastolic Blood Pressure, Mean Blood Pressure, Heart Rate, Respiratory Rate, Peak Inspiratory Pressure, Total PEEP Level, Tidal Volume Set, Tidal Volume Observed, Tidal Volume Spontaneous, SpO_2 (Pulse Oximetry), O_2 Flow, Temperature (°C), FiO_2 (Fraction of Inspired Oxygen), Cardiac Output
Laboratory Parameters	Albumin, Blood Urea Nitrogen, Bands (Immature Neutrophils), White Blood Cells, Bicarbonate, Creatine Kinase, Bilirubin, Creatine Kinase MB, Creatinine, Fibrinogen, Chloride, Lactate Dehydrogenase, Sodium, Magnesium, Potassium, Calcium (free), Lactate, pO_2 (Blood Gas), pCO_2 (Blood Gas), Hematocrit, Hemoglobin, pH (Blood Gas), Platelet Count, SO_2 (Blood Gas), Partial Thromboplastin Time, Glucose, Prothrombin Time (Quick), Troponin T, INR (Standardized Quick)

Table A.9.: Variables Structured Data based on Moor et al. (2019) [61]

B. Figures

B.1. M2: Boxplots Case vs. Controls (Section 5.2)

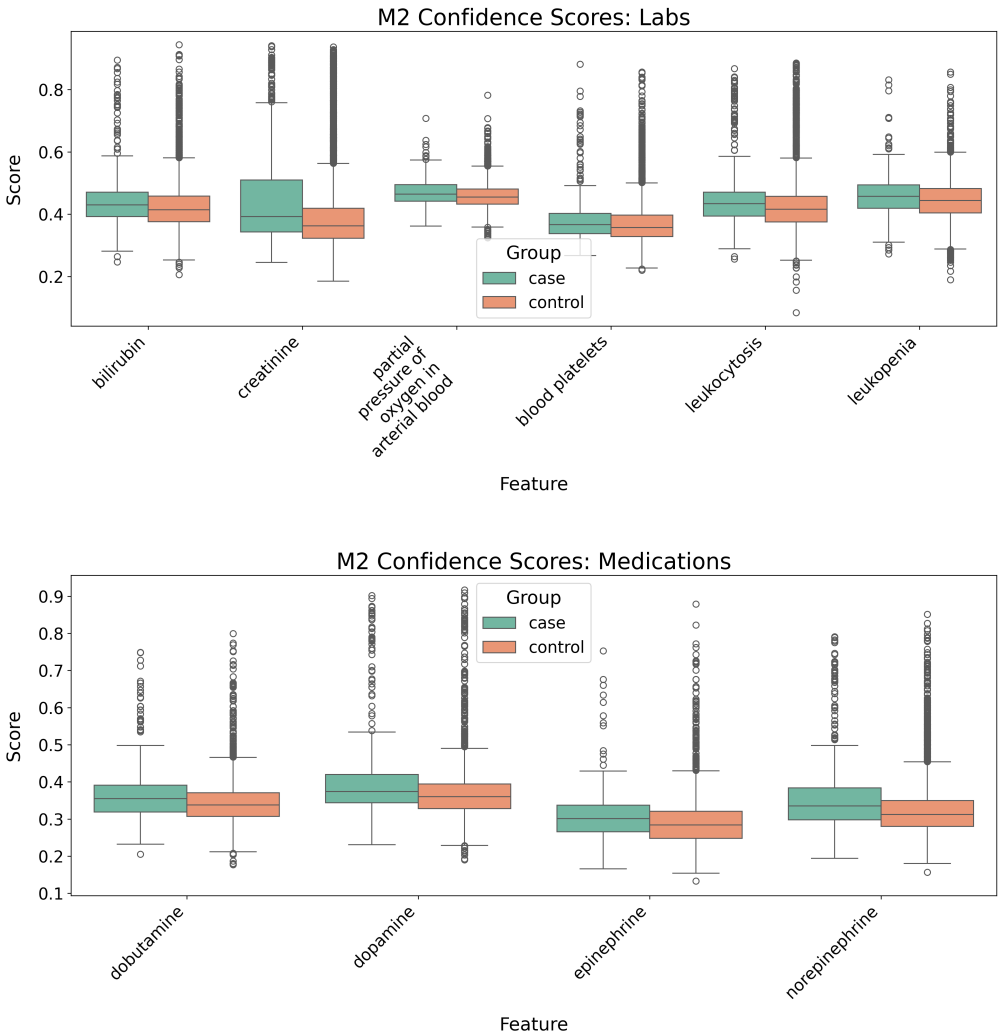


Figure B.1.: M2 model confidence score distributions for the **Laboratory** and **Medication** symptom categories. Each boxplot compares case and control groups across multiple features, highlighting differences in the model's score outputs between the two clinical populations.

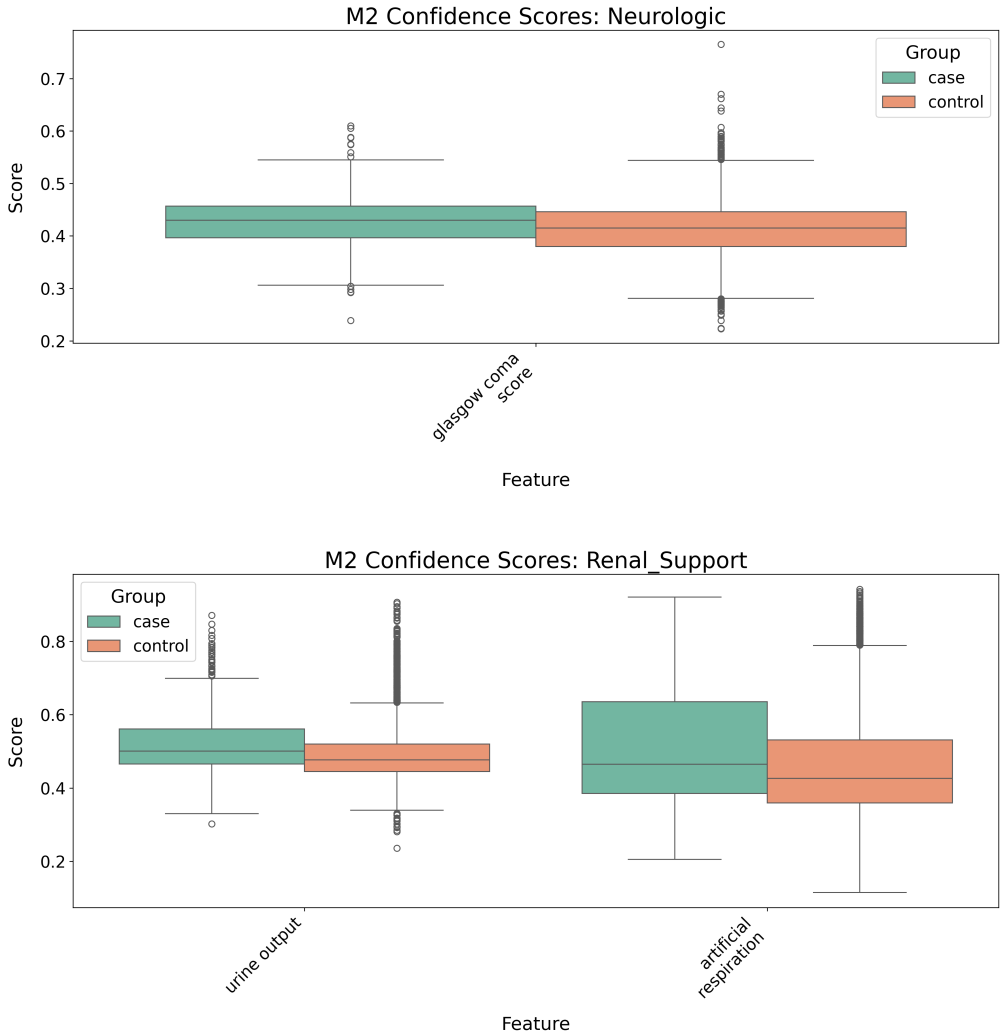


Figure B.2.: Comparison of M2 confidence scores between case and control groups for the **Neurologic** and **Renal Support** categories. The boxplots demonstrate the distribution of predicted symptom relevance, which may reflect neurological status and supportive care needs in patients.

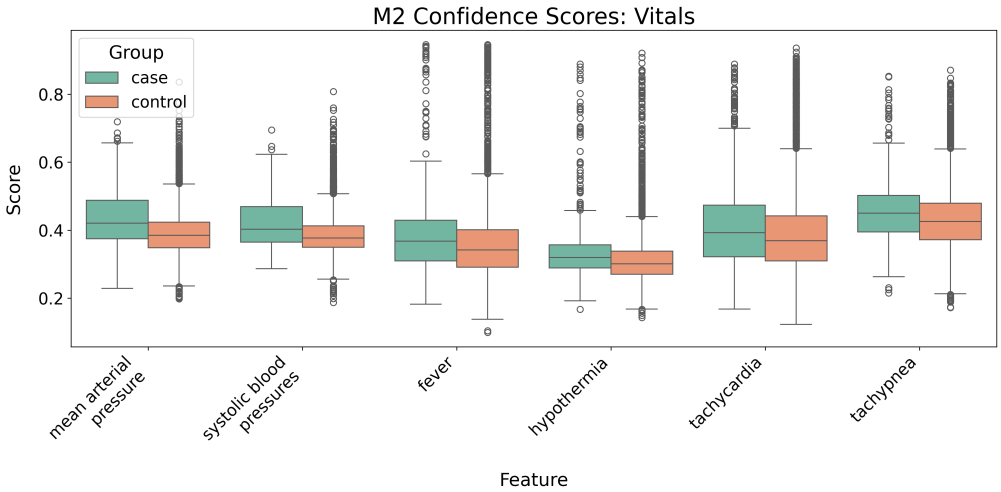
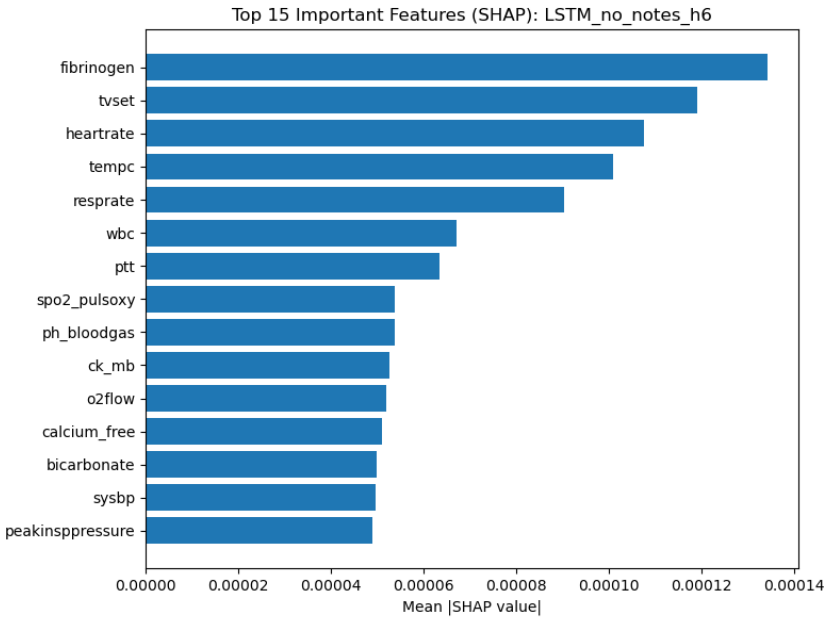
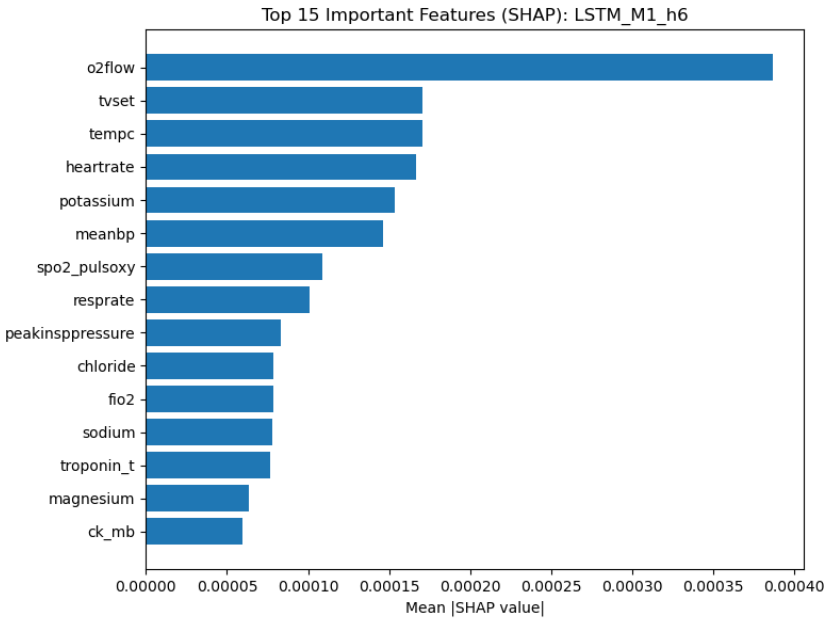


Figure B.3.: Boxplots of M2 confidence scores for the **Vital Sign** category, comparing case and control groups. This figure highlights the distribution of model predictions for dynamic physiological features such as blood pressure, temperature, and respiratory rate, which are critical indicators of clinical status.

B.2. M5: SHAP Values (Section 7.3)

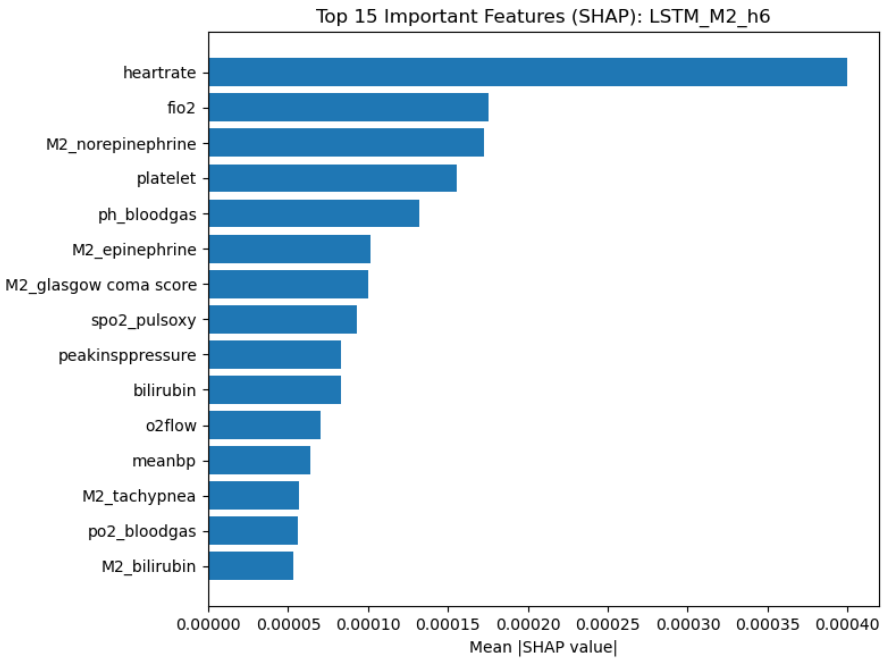


(a) Structured variables only

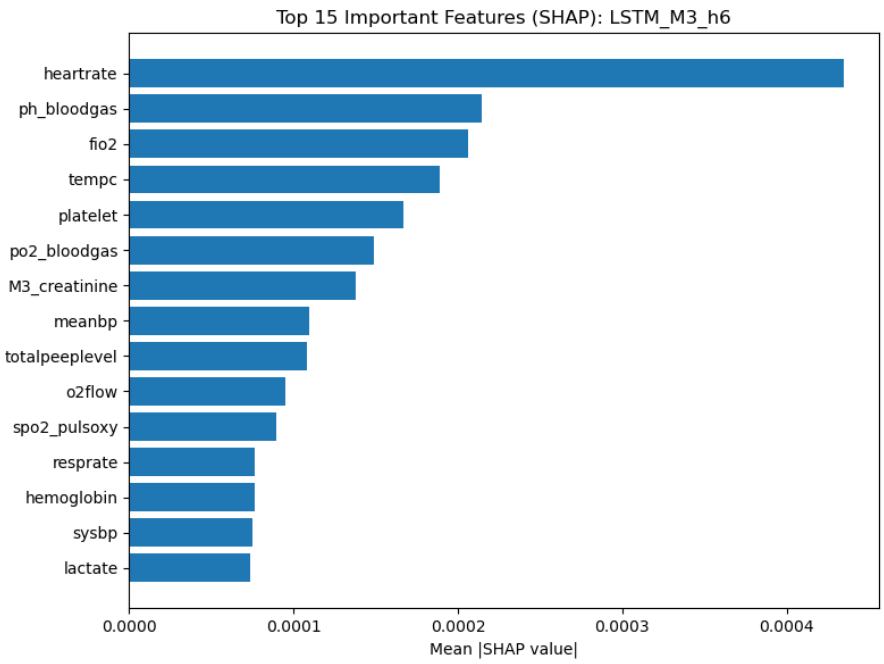


(b) M1 variables

Figure B.4.: Top SHAP values for BiLSTM models using the baseline model (structured data only) and rule-based method (M1) at a 6-hour horizon



(a) M2 variables



(b) M3 variables

Figure B.5.: Top SHAP values for BiLSTM models using LLM-derived (M2) and hybrid (M3) feature sets at a 6-hour horizon.

C. Digital Attachment

All scripts, preprocessing pipelines, and experimental configurations developed in this thesis are publicly available in the GitHub repository:

https://github.com/LilyIMV/clinical-notes-sepsis-llm

The repository is structured for reproducibility and transparency. It contains code for data processing and model training, along with documentation for running the experiments.

Repository Structure

```
.
|-- Makefile
|-- README.md
|-- environments/
|   |-- environment.yml
|   |-- environment_LSTM.yml
|   `-- environment_RAG.yml
|-- features.csv
|-- output/
|   |-- chunks/
|   |-- model/
|   |-- notes/
|   `-- query/
|-- slurm/
|   |-- M1.slurm
|   |-- M2.slurm
|   |-- M3.slurm
|   |-- M4.slurm
|   `-- logs/
```

```
src/
|-- model/
|   |-- config.py
|   |-- main.py
|   |-- model.py
|   |-- preprocessing/
|   |   |-- bin_and_impute.py
|   |   |-- collect_records.py
|   |   `-- main_preprocessing.py
|   `-- train.py
|-- nlp/
|   |-- M0_RAG.py
|   |-- M0_UMLS.py
|   |-- M0_preprocessing.py
|   |-- M1.py
|   |-- M2.py
|   |-- M2_postprocessing.py
|   |-- M3.py
|   `-- M3_postprocessing.py
`-- query/
    |-- cohort.sql
    |-- main_query.sql
    |-- match-controls.py
    |-- sofa-hourly.sql
    `-- ...
```

Description of Contents

The `Makefile` provides convenience commands for running experiments. The `README.md` contains installation and usage instructions. The file `features.csv` summarizes the extracted feature set, including regular expressions, prompts, and example matches.

The `environments/` directory contains YAML configuration files for reproducible software setup. These include a general environment, an environment for LSTM-based prediction, and one for retrieval-augmented generation (RAG).

The `output/` directory stores intermediate results and trained models. Subdirectories are used to organize text chunks, processed notes, model figures, and query outputs.

The `slurm/` directory contains job submission scripts (`M1.slurm`–`M4.slurm`) for executing the extraction pipelines on the HPC cluster, as well as a `logs/` subdirectory storing job outputs.

The `src/` directory contains the main codebase and is organized as follows:

- `model/`: scripts for predictive modeling, including preprocessing, configuration, and training. Parts of this code are based on Moor et al. (2019) [61].
- `nlp/`: feature extraction pipelines, including preprocessing (M0) and implementations of methods M1–M3.
- `query/`: SQL queries and helper scripts for cohort extraction, processing, and matching from MIMIC-III, partly based on Moor et al. (2019) [61].

Usage Instructions

To reproduce the results, the following steps can be followed:

1. Clone the repository.
2. Create the required software environment using the provided YAML files.
3. Run the SQL queries using `make query`.
4. Execute the extraction pipelines using `make M1`, `make M2`, and `make M3`.
5. Train and evaluate the prediction models using `make model`. To run a single pipeline, adjust `M4.slurm` and `config.py` accordingly.

Detailed instructions and usage examples are provided in the repository and in the `README.md`.

D. Experimental Details

All experiments were conducted on the high-performance computing server `gruenau10`, a Dell PowerEdge R740xd system configured for large-scale machine learning workloads. The server is equipped with two Intel Xeon Gold 6254 processors, each operating at 3.1 GHz, providing a total of 36 physical cores and 72 threads. This CPU infrastructure supported parallelized data preprocessing, rule-based feature extraction, and classical machine learning workflows.

The system is outfitted with 756 GB of RAM, enabling efficient handling of the MIMIC-III dataset and in-memory operations over large feature matrices and text corpora. For GPU-accelerated tasks—particularly those involving large language models and neural inference—the server is equipped with three NVIDIA Tesla A100 GPUs, each with 80 GB of high-bandwidth memory. These GPUs support mixed-precision computation and are compatible with modern deep learning frameworks via CUDA, OpenCL, and TensorFlow (version 12.4).

All experiments were implemented in Python using a modular and reproducible codebase. Two separate environments were configured to support different stages of the pipeline. The primary environment (`env`) was based on Python 3.9 and included core scientific libraries such as `pandas`, `scikit-learn`, `PyTorch 2.6.0`, `transformers`, and `ScispaCy`. This environment was used for model development, clinical feature extraction, and sepsis prediction tasks. It also supported GPU acceleration via `pytorch-cuda 12.1`.

A secondary environment (`env_RAG`), built on Python 3.11, was used for embedding-based retrieval and Retrieval-Augmented Generation (RAG). It included packages such as `InstructorEmbedding`, `sentence-transformers`, and `ChromaDB`, which were essential for similarity search and document indexing tasks.

Each method was executed via SLURM job scheduling, with larger workloads run in parallel to maximize resource utilization. Both environments are publicly available in the GitHub repository[1], ensuring reproducibility. Consistent chunking strategies and controlled batching were applied to guarantee deterministic outputs across runs.

[1] `https://github.com/LilyIMV/clinical-notes-sepsis-llm/tree/main`

Selbstständigkeitserklärung

Ich erkläre hiermit, dass ich die vorliegende Arbeit selbständig verfasst und noch nicht für andere Prüfungen eingereicht habe. Sämtliche Quellen einschließlich Internetquellen, die unverändert oder abgewandelt wiedergegeben werden, insbesondere Quellen für Texte, Grafiken, Tabellen und Bilder, sind als solche kenntlich gemacht. Mir ist bekannt, dass bei Verstößen gegen diese Grundsätze ein Verfahren wegen Täuschungsversuchs bzw. Täuschung eingeleitet wird.

Berlin, den February 11, 2026 .....